Dynamic Nonlinear Seismic Response Analysis of the RC Upright Construction Method

June 22, 2005

Research Report from a Prominent Institution

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1. Overview

The purpose of this analysis is to analytically verify the structural integrity of a two-story building utilizing the **RC Upright Construction Method** (reinforced concrete structure) when subjected to major earthquakes corresponding approximately to **Japan Meteorological Agency Seismic Intensity Levels 6 to 7**.

The **RC Upright Construction Method** is an entirely new construction system consisting of a reinforced concrete structure with a **mat foundation** (450 mm or greater in thickness) and **one-directional exterior and partition walls** (300 mm or greater in thickness), combined with **timber floors, walls, and roof structures constructed using the 2×4 method**.

The RC Upright Construction Method possesses outstanding characteristics in **adaptability, fire resistance, seismic performance, condominium ownership compatibility, sound insulation, renovation flexibility, and architectural design freedom**. A patent application for this construction system is currently being filed by your company.

In this study, the two-story RC Upright Construction Method building is modeled using **three-dimensional thick-shell elements incorporating reinforcing bars**, and a **nonlinear dynamic seismic response analysis** is performed to evaluate its structural integrity under earthquake loading.

The analysis is conducted using the **general-purpose finite element analysis program ABAQUS Version 6.5**.

2. Analysis Conditions

2.1 Structure Analyzed

Figure 1 shows an example of a building constructed using the **RC Upright Construction Method** (Takashima Apartment Building). Figure 2 presents an overview of the **two-story RC Upright Construction Method building**.

Figure 3 shows the drawings of the analytical model used in this study (two-story RC Upright Construction Method building). As illustrated in Figure 3, the analytical model consists of a single reinforced concrete structural system comprising a **mat foundation, two exterior walls, and two partition walls**.

The principal structural dimensions of the analytical model are as follows:

- Wall dimensions: **5,400 mm (height) × 300 mm (thickness) × 10,310 mm (width)**
- Foundation slab dimensions: **13,500 mm × 10,310 mm × 500 mm (thickness)**

The reinforcing bars are **SD295 grade steel** with **lap-splice connections**.

The arrangement of the wall main reinforcement (vertical reinforcement) is as follows:

- **D16 @160W** for the portion above **GL + 2150 mm**
- **D13 and D16 @80W** for the portion below **GL + 2150 mm**

The horizontal reinforcement is arranged as:

- **D10 @100W**

The effective depth of the wall is:

- **d = 240 mm**

For the foundation slab, the main reinforcement is arranged as:

- **D13 and D16 @160W**

The stirrups are arranged as:

- **D13 @190W**

The effective depth of the foundation slab is:

- **d = 410 mm**

The unit system adopted in this analysis is the **SI system (N, mm, sec)**.

2.2 Analysis Code

The general-purpose finite element analysis program **ABAQUS** is used in this study.

The ABAQUS finite element analysis system consists of:

- **ABAQUS/Standard** (general-purpose finite element analysis program)
- **ABAQUS/Explicit** (explicit dynamic finite element analysis program)
- **ABAQUS/CAE** (a complete ABAQUS operating environment including pre-processing and post-processing tools)

ABAQUS/Standard is a general-purpose finite element program developed for advanced engineering analyses and practical applications.

At present, ABAQUS is one of the most widely used and well-established analysis codes both in Japan and internationally. It has been extensively applied in a wide range of engineering fields, including:

- Automotive Engineering
- Mechanical Engineering
- Aerospace Engineering
- Nuclear Engineering
- Heavy Industry
- Building and Civil Engineering
- Chemical Engineering

ABAQUS is widely used for the verification of **structural safety** and **structural integrity** in structural design and engineering practice, and is regarded as a highly reliable analysis code.

ABAQUS features:

- An extensive element library
- A large collection of verification and benchmark examples
- Advanced automatic increment control capabilities

These features make ABAQUS particularly suitable for complex nonlinear structural analyses and advanced engineering applications.

2.3 Analytical Model

(1) Reinforced Concrete Foundation and Walls

As an initial stage of this study, the two-story RC Upright Construction Method building shown in Figure 3 is modeled using **three-dimensional thick-shell elements**.

The concrete portions of the walls and foundation slab are modeled using three-dimensional thick-shell elements located at the mid-surface of the wall and slab thicknesses. The reinforcing bars are modeled using the ABAQUS reinforcement element function (***REBAR LAYER data**). By combining these models, the walls and foundation slab are analyzed as a **reinforced concrete structure**.

In the three-dimensional shell model, only the **main reinforcing bars (vertical reinforcement)** and **horizontal reinforcing bars** can be modeled explicitly. However, the **hoop reinforcement** in the thickness direction, which connects the horizontal reinforcing bars, cannot be modeled.

In the case of the two-story RC Upright Construction Method building, the wall height is approximately **5,400 mm**, while the wall thickness is **300 mm**, resulting in a relatively slender wall configuration. Therefore, the expected failure mode during an earthquake is considered to be **flexural failure**. Consequently, the contribution of hoop reinforcement, which primarily resists wall buckling and shear failure, is considered to be relatively small.

In addition, although wall segments exist around openings such as windows and doors, these wall segments are conservatively neglected in the present analysis. The openings are modeled as full-height vertical slits extending over the entire wall height.

(2) Timber Floors, Walls, and Roof Structure

The timber floor, wall, and roof components are not explicitly modeled in the present analysis. However, the masses of these components are included as additional loads.

The additional loads corresponding to the timber floors, walls, and roof structures are described in **Section 2.6**.

(3) Soil

The soil is modeled using **horizontal and rotational soil springs**.

The soil spring constants used in the analysis are described in **Section 2.7**.

2.4 Material Properties

(1) Concrete

Since **ABAQUS/Standard Version 6.3**, a new concrete constitutive model, the **Concrete Damaged Plasticity Model**, has been introduced to improve the accuracy of structural analyses of reinforced concrete structures. In this study, the **Concrete Damaged Plasticity Model** is adopted as the constitutive model for concrete. A detailed description of this concrete constitutive model is provided in **Appendix 4**.

The concrete used for the walls and foundation of the two-story RC Upright Construction Method building is normal-weight concrete with a specified design compressive strength of:

Fc = 21 N/mm²

In this analysis, the concrete of the walls and foundation slab is modeled as a nonlinear material that accounts for concrete cracking.

The material properties used for the concrete are as follows:

- Young's Modulus: **EC = 21,700 N/mm²**
- Poisson's Ratio: **ν = 0.167**
- Compressive Strength: **σU = 21 × 0.85 / 1.3 = 13.731 N/mm²**

- Ultimate Compressive Strain: ε **CU' = 0.0035**
- Tensile Strength: σ **T = 3.882 N/mm²**

The tensile strength was determined as:

ftk = 0.23fck^(2/3) + Bond Strength 0.28fck^(2/3)

Figure 2.26 shows the uniaxial stress–strain relationship for concrete.

As shown in the figure:

- The compression side is modeled as a **bilinear relationship**.
- The tension side is modeled as a **trilinear relationship**.

The tension stiffening behavior that governs the post-cracking response is shown in **Figure 2.26(b)**.

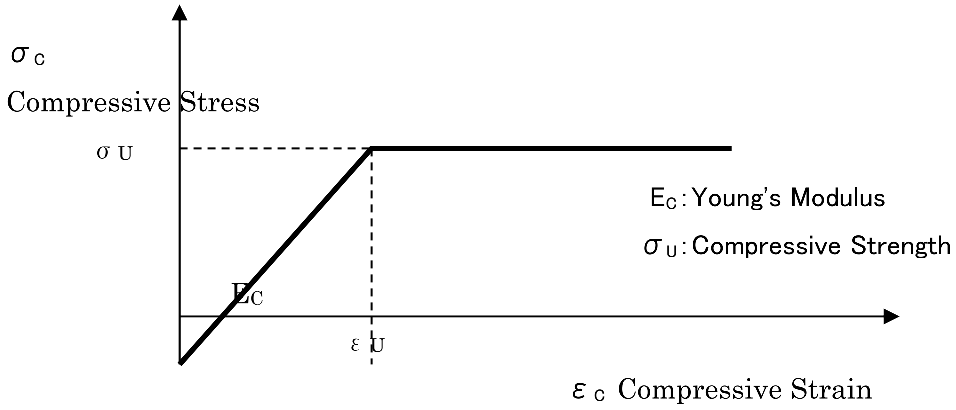
Figure 2.27 illustrates the cyclic characteristics of the Concrete Damaged Plasticity Model.

For the ABAQUS Concrete Damaged Plasticity Model used in this analysis, the following parameters are adopted:

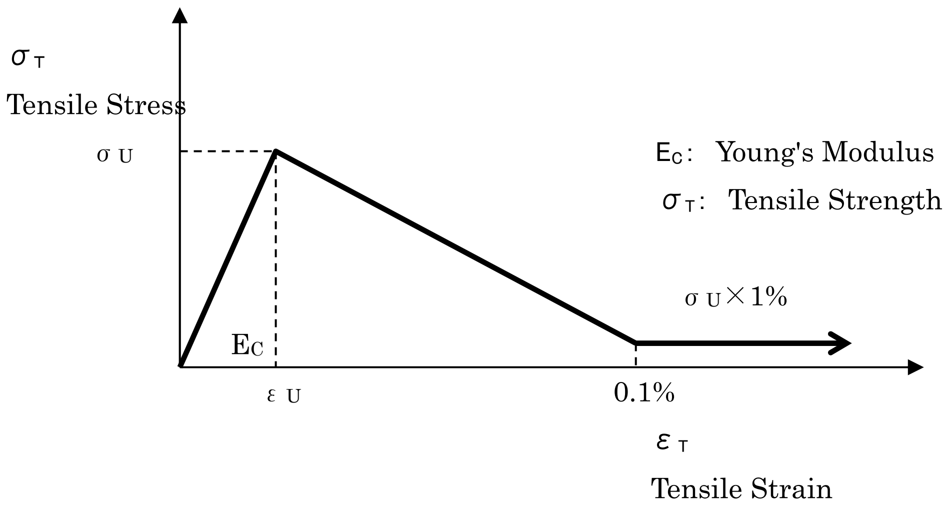
- Dilation Angle: ϕ **= 35°**
- Viscosity Parameter: μ **= 1.0 × 10^{-5}**
- Compression Stiffness Recovery Factor: **WC = 1.0** (default value)
- Tension Stiffness Recovery Factor: **WT = 0.0** (default value)

The concrete material properties are defined in ABAQUS using the following options:

```
*CONCRETE DAMAGED PLASTICITY
*CONCRETE COMPRESSION HARDENING
*CONCRETE TENSION STIFFENING, TYPE=STRAIN
*CONCRETE TENSION DAMAGE, TYPE=STRAIN
```



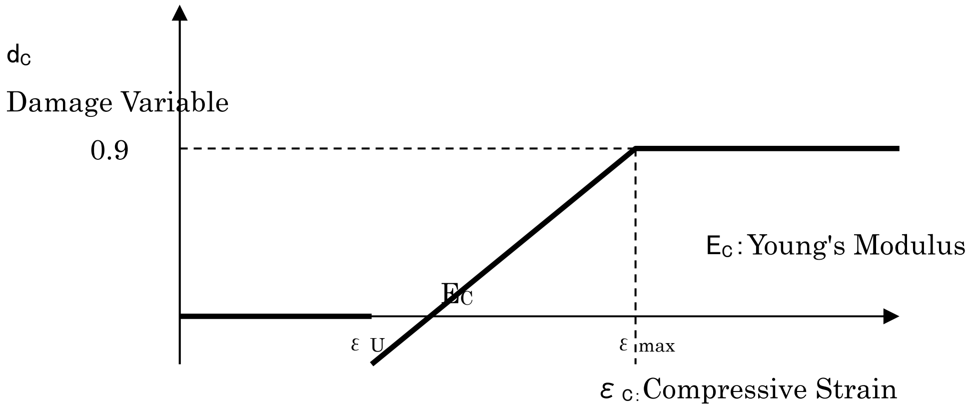
(a) Compression Side (Bilinear Model)



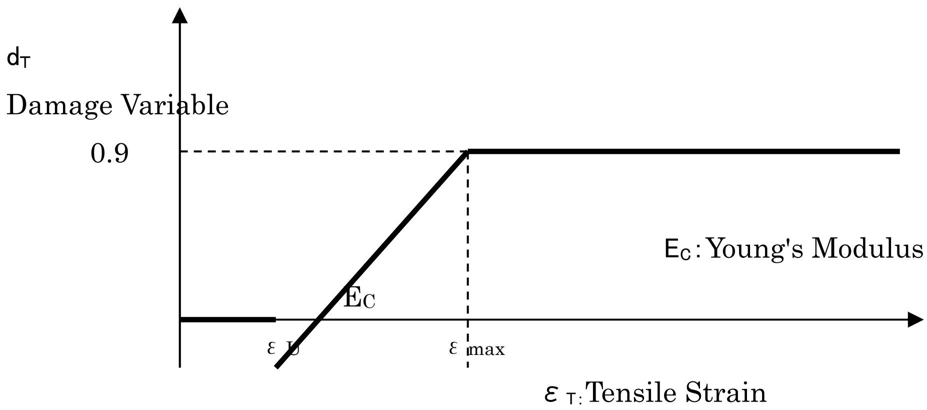
(b) Tension Side (Trilinear Model)

図 2.26　コンクリートの応力－ひずみ関係（1 軸）

Figure 2.26 Stress–Strain Relationship of Concrete (Uniaxial Loading)



(a) Compression Side



(b) Tension Side

Figure 2.27 Damage Variable–Strain Relationship of Concrete (Uniaxial Loading)

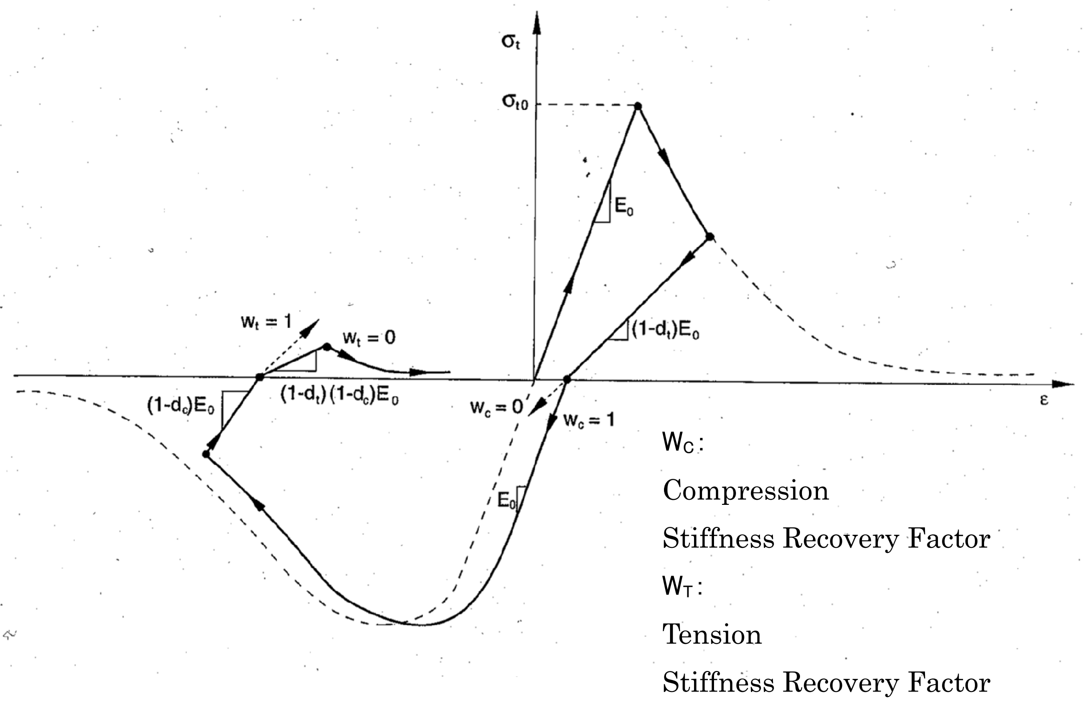


Figure 2.28 Cyclic Behavior of the ABAQUS Concrete Damaged Plasticity Model

(2) Reinforcing Steel

The reinforcing bars used in this study are **SD295 grade steel with lap-splice connections**.

The material properties adopted for the reinforcing bars, including **Young's modulus (E)**, **Poisson's ratio**, **yield stress**, and the **nonlinear stress-strain relationship**, are as follows.

Since no prestressing force is applied to the reinforcing bars in the two-story RC Upright Construction Method building, **prestressing effects are not considered in this analysis**.

- **Young's Modulus:**
  ES=200,000 N/mm2E_S = 200,000 ¥, ¥text{N/mm}^2ES=200,000N/mm2
- **Poisson's Ratio:**
  ν=0.3¥nu = 0.3 ν=0.3

- **Stress–Strain Relationship:**
  The reinforcing steel is modeled using a **bilinear stress–strain relationship**, as shown in **Figure 2.29**.
- **Yield Stress:**
  σy=295 N/mm2¥sigma_y = 295 ¥, ¥text{N/mm}^2 σy=295N/mm2

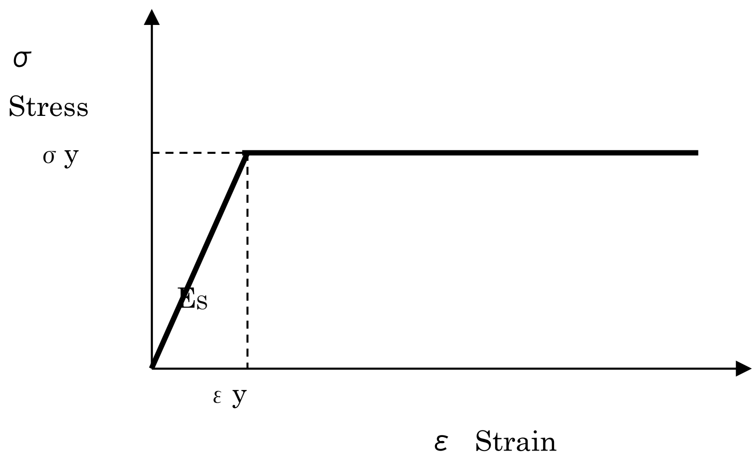


Figure 2.29 Stress–Strain Relationship of Reinforcing Steel

2.5 Boundary Conditions

The RC Upright Construction Method utilizes a **mat foundation**. The boundary conditions adopted in the seismic response analysis are as follows.

Under the initial condition (self-weight loading), the vertical displacement of the bottom surface of the mat foundation is supported by **vertical springs**.

During the seismic response analysis, the bottom surface of the mat foundation is supported by **horizontal soil springs** and **rotational soil springs**.

For the **modal analysis (natural vibration analysis)** of the building, the bottom surface of the mat foundation is assumed to be **fixed**.

2.6 Loading Conditions

(1) Input Earthquake Motion

The criteria for the selection of **horizontal input earthquake motions** specified in Reference (11), Methodology for Performance Evaluation of Buildings Using Time-History Response Analysis published by the **Japan Building Center (JBC)** for high-rise buildings, are outlined below.

(1) Earthquake motions having the acceleration response spectrum specified for the engineering bedrock in Notification No. 4, Item (a) and generated by appropriately considering the amplification effects of the surface soil layers at the construction site (hereinafter referred to as "Notification Waves") shall be used as the design input earthquake motions.

In this case, three or more earthquake records shall be used that satisfy the requirements specified in Notification No. 4, Item (a), including duration and other relevant conditions, and that have been generated with appropriate consideration of phase characteristics.

(2) In accordance with the proviso of Notification No. 4, Item (a), when simulated earthquake motions for the construction site (hereinafter referred to as "Site-Specific Waves") are appropriately developed based on the distribution of active faults surrounding the site, fault rupture models, historical seismic activity, ground conditions, and other relevant factors, such site-specific waves may be used as the design input earthquake motions in place of the earthquake motions corresponding to extremely rare seismic events specified in the Notification Waves described above.

In this case, three or more earthquake records generated with appropriate consideration of phase characteristics and other relevant factors shall be used. When Notification Waves are used together with Site-Specific Waves, the total number of records shall be three or more.

(3) In either case described in (1) or (2) above, the following earthquake motions shall also be used as design input earthquake motions in order to verify the appropriateness of the generated earthquake records.

Specifically, from among representative recorded earthquake motions observed in the past, three or more records shall be selected appropriately, taking into consideration the characteristics of the construction site and the building.

Earthquake motions generated by scaling the maximum velocity amplitude of these records to:

- 250 mm/s shall be regarded as earthquake motions that occur rarely, and
- 500 mm/s shall be regarded as earthquake motions that occur extremely rarely.

Furthermore, the above values of maximum velocity amplitude may be multiplied by the seismic zone coefficient Z specified in Article 88, Paragraph 1 of the Enforcement Order.

4.4.1 Selection of Horizontal Input Earthquake Motions

As described above, for the performance evaluation of high-rise buildings using time-history response analysis, it is necessary to use at least **three code-specified artificial ground motions, three site-specific ground motions, and three recorded**

earthquake ground motions as design input earthquake motions. The code-specified and site-specific ground motions must be generated by the designer in accordance with the *Technical Guidelines (Draft) for the Generation of Design Input Ground Motions* presented in **Reference (10)**.

However, the subject structure of this study is a **two-story building**, and the structural performance of low-rise buildings is generally evaluated using conventional static seismic force procedures.

For the nonlinear dynamic earthquake response analysis conducted in this study, one earthquake ground motion record was selected from the twelve representative recorded earthquake motions shown in **Figures 2.14 through 2.25** and **Appendix 2**. These records were identified by the High-Rise Building Structural Evaluation Committee in *Building Letter*, June 1986 issue, published by the **Building Center of Japan**, entitled *"Earthquake Motions for Dynamic Analysis of High-Rise Buildings."* Among these records, the **El Centro NS ground motion** (**peak acceleration: 341.7 cm/s²; peak velocity: 33.5 cm/s**) was adopted because it is regarded as one of the most reliable recorded earthquake motions currently available.

The earthquake motion was applied in a single horizontal direction corresponding to the **weak-axis direction of the reinforced concrete walls**.

In addition, Appendix 3 presents the Design Response Spectra for Horizontal Seismic Motions.

These spectra are referenced from **Reference (10): Building Research Institute, Ministry of Construction and Japan Building Center, Technical Guidelines (Draft) for the Preparation of Design Input Earthquake Motions, Commentary Edition, March 1992.**

In the **design velocity response spectra (Level 1 and Level 2)** shown in Appendix 3, a velocity response value of **100 kine** corresponds approximately to an **acceleration response spectrum of 1000 gal**.

The thick solid line shown in the figures represents the **velocity response spectrum of the design artificial earthquake wave B(T)**.

The subsequent pages present the:

- Acceleration time histories
- Velocity time histories
- Displacement time histories

of the design artificial earthquake waves **B(T), Level 1, and Level 2**.

The maximum accelerations of the artificial earthquake wave B(T) are:

- **207.3 gal (Level 1)**
- **355.7 gal (Level 2)**

However, in the corresponding design velocity response spectra, the maximum values are:

- **50 kine (Level 1, equivalent to approximately 500 gal)**
- **100 kine (Level 2, equivalent to approximately 1000 gal)**

The design velocity response spectra (Level 1 and Level 2) also include the velocity response spectra of the **El Centro NS record**, scaled to:

- **20 kine for Level 1**
- **50 kine for Level 2**

These spectra are indicated by the **dashed lines with yellow markers** in the figures.

The **El Centro NS earthquake record** used in the present seismic response analysis, having:

- Maximum Acceleration = **341.7 gal**
- Maximum Velocity = **33.5 kine**

is considered to correspond to an acceleration response spectrum with a maximum value of approximately **700 gal**.

For reference, the **acceleration response spectra (h = 5%)** and **velocity response spectra (h = 5%)** of the **1995 Hyogo-ken Nanbu Earthquake** and the **2004 Niigata Chuetsu Earthquake**, obtained from the website of the **Disaster Prevention Research Institute, Kyoto University (Iwata Laboratory)**, are also presented.

As shown in the figures, the **Niigata Chuetsu Earthquake** exhibits significantly larger response spectra than the **Hyogo-ken Nanbu Earthquake**, with:

- Maximum Acceleration Response Spectrum: **6,000 gal (h = 5%)**
- Maximum Velocity Response Spectrum: **600 kine (h = 5%)**

These values demonstrate the exceptionally large seismic demand generated by the Niigata Chuetsu Earthquake compared with that of the Hyogo-ken Nanbu Earthquake.

(2) Additional Masses of Timber Components

In the present analysis, the timber components, including the walls, floors, and roof, are not explicitly modeled. However, their masses are incorporated into the analytical model as additional masses.

The masses of the timber walls and floors are assigned to the reinforced concrete walls at the floor level.

The masses of the timber roof and floor systems are assigned at the roof center-of-gravity level and at the top of the reinforced concrete walls.

The values and locations of these additional masses are presented in **Appendix 1**.

2.7 Soil Springs

In this analysis, the bottom surface of the mat foundation is modeled using **horizontal and rotational soil springs** in order to account for the **dynamic soil-structure interaction** between the ground and the structure.

The soil spring constants are determined based on the **vibration admittance theory** presented in **Reference (5), JEAG 4601 – Seismic Design Guidelines for Nuclear Power Facilities**, pp. 318–321, in which the soil spring constants for a rectangular foundation mat are defined.

However, **static soil springs**, rather than dynamic soil springs, are adopted in the present analysis.

(1) Assumed Soil Conditions

Since buildings are generally constructed on a variety of ground conditions, the soil properties adopted in this analysis are based on those presented in **Reference (5), JEAG 4601, p.233**.

The assumed ground conditions are as follows:

- **Alluvial Sand Layer**
- Shear Wave Velocity: **VS = 150 m/s**
- Standard Penetration Test (SPT) N-Value: **N = 15**
- Poisson's Ratio: **ν = 0.4**
- Unit Weight: **ρ = 1.8 t/m³**
- Shear Modulus $G=\rho V_S^2/g=413(kgf/cm^2)$
- Young's Modulus $E=2(1+\nu)G=1157(kgf/cm^2)$

(2) Vertical Soil Spring Constant (Self-Weight Condition)

For the self-weight analysis, the foundation is supported by **vertical soil springs**.

The vertical soil spring constant for the rectangular mat foundation is determined based on the **vibration admittance theory** presented in **Reference (5), JEAG 4601, pp. 318–321**, as follows:

- **Vertical Soil Spring Constant**

$$K_V=G\sqrt{(A)}\times a_V/(1-\nu)$$
$$=40.5\times\sqrt{(10310\times13500)}\times2.15/(1-0.4)$$
$$\therefore\ K_V=1.713\times10^6(N/mm)$$

(3) Horizontal and Rotational Soil Spring Constants (Seismic Condition)

For the seismic response analysis, the foundation is supported by **horizontal and rotational soil springs** in order to account for the **dynamic soil-structure interaction** between the ground and the structure.

The horizontal and rotational soil spring constants for the rectangular mat foundation are determined based on the **vibration admittance theory** presented in **Reference (5), JEAG 4601, pp. 318–321**, and are calculated as follows.

- Horizontal Soil Spring Constant

$$K_H = G\sqrt{(A)} \times a_H$$
$$= 40.5 \times \sqrt{(10310 \times 13500)} \times 2.75$$
$$\therefore\ K_H = 1.314 \times 10^6 (N/mm)$$

- Rotational Soil Spring Constant

$$K_R = \pi G Z y \times a_R / (1 - \nu)$$
$$= \pi \times 40.5 \times (10310 \times 13500^2/6) \times 2.3 / (1-0.4)$$
$$\therefore\ K_H = 1.527 \times 10^{14} (N \cdot mm/rad)$$

2.8 Damping Characteristics

For the material damping characteristics adopted in this analysis, **Rayleigh damping** (indicated by the blue line in Figure 2.30) is used, and the ABAQUS ***DAMPING** option is employed.

The damping coefficients α and β are determined based on the modal analysis results presented in Chapter 3. Specifically, the coefficients are established by considering:

- The **first natural frequency,** f_1, which has the largest participation factor and corresponds to the **first bending vibration mode of the partition wall** (indicated by the red circle in the figure), and
- The **tenth natural frequency,** f_{10}, which has a relatively large participation factor and corresponds to the **second bending vibration mode of the partition wall** (indicated by the yellow circle in the figure).

The damping ratio **h** is selected based on the recommendations given in **Reference (5), JEAG 4601**, which states that the damping ratio for reinforced concrete structures generally ranges from **0.02 to 0.05**.

In the present analysis, a damping ratio of:

h=0.03h = 0.03h=0.03

is adopted in order to account for the nonlinear behavior of the reinforced concrete structure.

$[C]= \alpha[m]+\beta[k]$

ここに、$[C]$：Damping Matrix

$[m]$：Mass Matrix

$[k]$：Stiffness Matrix

α，β：Damping Coefficient

$\alpha = h\times 2\omega_1\times\omega_{10}/(\omega_1+\omega_{10})$

$= 0.03\times(2\pi\times 3.126)\times(2\pi\times 9.158)/((2\pi\times 3.126)+(2\pi\times 9.158))$

$\therefore \alpha = 0.8786$

$\beta = h\times 2/(\omega_1+\omega_{10})$

$= 0.03\times 2/((2\pi\times 3.126)+(2\pi\times 9.158))$

$\therefore \alpha = 7.7738\times 10^{-4}$

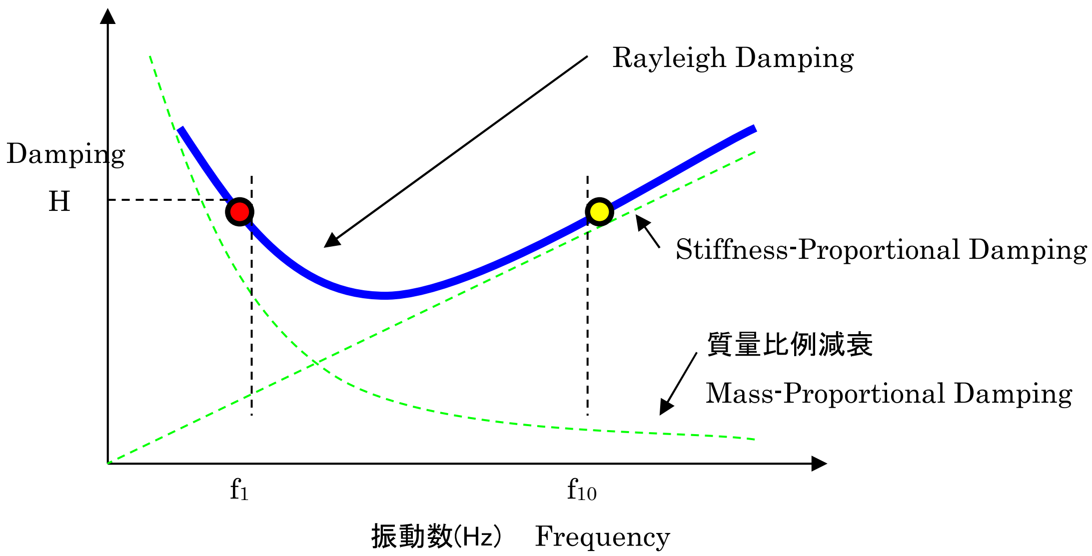


図 2.30　レーリー減衰

Figure 2.30 Rayleigh Damping

2.9 Analysis Method

In this study, a **static self-weight analysis** followed by a **nonlinear dynamic seismic response analysis** is performed for the two-story RC Upright Construction Method building.

A **time-history nonlinear dynamic analysis** based on the **direct integration method** is employed.

The integration scheme used in the general dynamic analysis procedure of **ABAQUS/Standard** is the **Hilber–Hughes–Taylor (HHT) method**, which is an extension of the trapezoidal rule.

The Hilber–Hughes–Taylor method is an **implicit integration method**, requiring the solution of the nonlinear dynamic equilibrium equations at each time increment.

The nonlinear equations are solved iteratively using the **Newton–Raphson method**.

For each time interval, Δt (s), of the input earthquake motion used in this analysis, an **implicit dynamic time integration** is performed using the **automatic time incrementation scheme of ABAQUS**, and converged solutions are obtained.

2.10 Analysis Cases

The analysis consists of **one case**, comprising a **static self-weight analysis** followed by a **nonlinear dynamic seismic response analysis** using a single input earthquake motion record, namely the **El Centro NS earthquake wave**.

In addition, a **modal analysis** is performed to evaluate the dynamic characteristics and vibration properties of the RC Upright Construction Method building.

2.11 Analysis Outputs

The following analytical results of the RC Upright Construction Method building are output and used as the basis for evaluating the structural integrity of the building during earthquakes.

- **Natural Vibration Mode Shapes and Animations** (showing the actual wall and slab thicknesses)
- **Deformed Shapes and Animations at the Time of Maximum Acceleration** (showing the actual wall and slab thicknesses)
- **Acceleration Contour Plots at the Time of Maximum Acceleration** (showing the actual wall and slab thicknesses)
- **Section Force Contour Plots at the Time of Maximum Acceleration** (showing the actual wall and slab thicknesses)
- **Equivalent Plastic Strain Contour Plots and Animations** (showing the actual wall and slab thicknesses)
- **Concrete Cracking Damage Contour Plots** (showing the actual wall and slab thicknesses)
- **Acceleration Time-History Plots at the Top of the Exterior Wall and at the Second-Floor Level of the Exterior Wall**
- **Acceleration Time-History Plots at the Top of the Partition Wall and at the Second-Floor Level of the Exterior Wall**
- **Relative Displacement Time-History Plots between the Top of the Exterior Wall and the Slab**
- **Relative Displacement Time-History Plots between the Top of the Partition Wall and the Slab**
- **Maximum Response Ductility Factor**

Key Features of the RC Upright Construction Method

Construction Cost

"We want to create a building with outstanding architectural design. However, we also want to keep construction costs under control."

This is a common objective shared by many property owners considering rental housing investments.

The RC Upright Construction Method was developed to achieve exactly that goal.

The construction examples completed to date have been delivered at costs comparable to those of medium-grade timber residential buildings. Furthermore, through continued improvements in construction methods, engineering development, and project execution processes, we aim to achieve even greater cost reductions in the future.

At the same time, higher specifications and enhanced features can be incorporated through optional upgrades. By working closely with each owner's business plan and investment strategy, we propose attractive, high-value buildings tailored to their specific needs.

Sound Insulation

One of the most common problems in apartment buildings is noise-related disputes.

To protect the privacy and comfort of residents, the RC Upright Construction Method incorporates a highly effective sound-insulation structure into the building itself.

The 300 mm thick reinforced concrete walls provide excellent sound insulation performance, significantly reducing the transmission of noise between neighboring units.

This contributes to a more comfortable living environment and helps residents build positive relationships with their neighbors. In turn, it also provides substantial benefits to property owners by enhancing tenant satisfaction and the long-term value of the property.

Flexibility

Trends are constantly changing, and the requirements placed on buildings will continue to evolve over the next 10 or 20 years. In an era of such rapid change, adaptability and ease of renovation are essential considerations.

By retaining the reinforced concrete (RC) structural walls that form the building' s permanent foundation, while allowing the other elements—such as walls, floors, and roofs—to be redesigned and renewed in accordance with changing lifestyles and architectural trends, the building can continue to meet market demands and attract occupants for generations to come.

Impact of Full Renovation Capability on Asset Value

In rental apartment buildings, once residents are living in their units, renovation work can be difficult to undertake. Property owners cannot simply require tenants to move out in order to carry out construction.
The RC Upright Construction Method is designed with a high degree of adaptability, allowing for partial renovations and modifications without requiring the entire building to be rebuilt. This flexibility enables rental housing to respond effectively to changing market demands, lifestyles, and design trends over time. As a result, it provides a sustainable and future-ready housing solution for both owners and residents.

Figure 2.1 Key Features of the RC Upright Construction Method (1/2)

Advantages of Condominium Ownership

Many property owners build rental housing with the intention of eventually passing the land and buildings on to their children as part of their estate planning.
The RC Upright Construction Method, which allows for sectional ownership, is particularly well suited to such long-term asset management strategies. By enabling ownership to be divided by unit, it provides greater flexibility for inheritance planning and asset succession, making it an attractive option for owners who wish to preserve and transfer their property to future generations.

Reduced Risk of Termite Damage

Many conventional wood-frame and 2×4 construction buildings are structurally vulnerable to termite damage. Once termites infest a building, they can seriously affect critical structural components such as load-bearing columns and structural walls.

In contrast, the primary structural framework of the RC Upright Construction Method is made of reinforced concrete (RC). Therefore, even if termite damage occurs in non-structural components, it is unlikely to compromise the building's structural integrity. This provides greater durability, safety, and long-term reliability for property owners.

Fire Safety

Buildings constructed using wood-frame, 2×4, or prefabricated construction methods are generally more susceptible to the spread of fire than reinforced concrete (RC) buildings.

In the RC Upright Construction Method, the partition walls between adjacent dwelling units as well as the exterior walls are constructed with 30 cm thick reinforced concrete walls. This robust fire-resistant design is expected to provide superior fire protection performance, even compared with conventional RC buildings, helping to reduce the risk of fire spread and enhance the safety of residents and property owners alike.

Seismic Resistance

The RC Upright Construction Method utilizes a combination of heavy-duty reinforced concrete (RC) shear walls and lightweight 2×4 walls, resulting in a structure that is lighter than conventional RC buildings while maintaining excellent structural performance. (Required soil bearing capacity: $5\ t/m^2$).

In an actual completed project, a dynamic seismic performance evaluation was conducted. The results demonstrated that the building could withstand the Nōbi Earthquake (Magnitude 7.2) scenario anticipated by the City of Nagoya.

This advanced housing system offers significant peace of mind not only for property owners but also for tenants, who increasingly value disaster-resistant housing. Designed to address the challenges posed by natural disasters while maintaining flexibility and long-term asset value, the RC Upright Construction Method represents a next-generation approach to residential construction.

We are proud to offer this innovative and forward-looking building solution.

Design

In recent years, interest in residential design has grown significantly, to the point where it is frequently featured in television programs and magazines. We are now living in an era in which people seek to express their lifestyles and personal values through the homes they choose to live in.
To meet the expectations of these increasingly design-conscious residents, we aim to continuously introduce innovative architectural styles and living environments through collaboration with a wide range of designers. By combining flexibility, functionality, and outstanding design, we strive to create housing that remains attractive and relevant as lifestyles and trends continue to evolve.

Figure 2.1 Key Features of the RC Upright Construction Method (2/2)

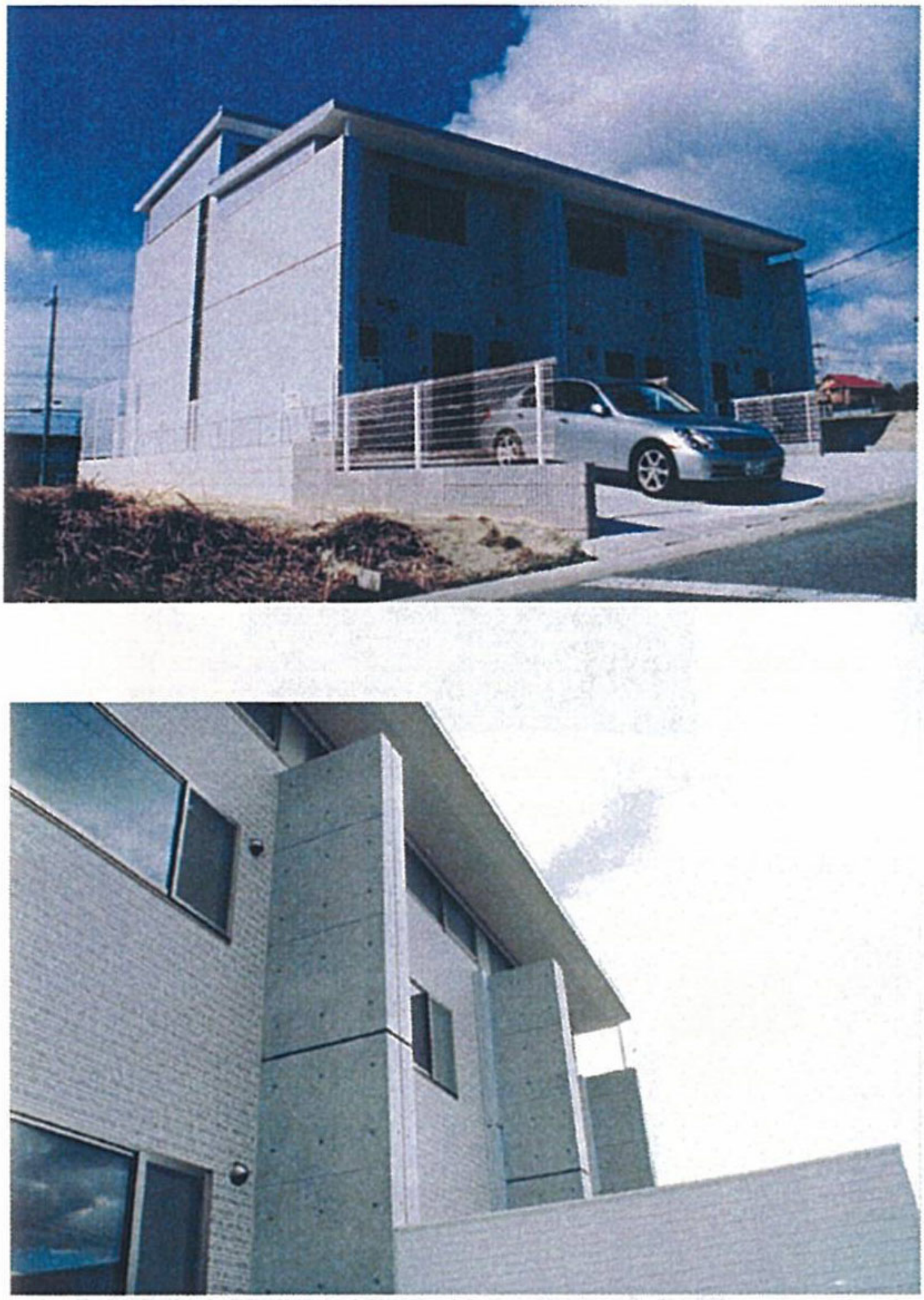


Figure 2.1 RC Upright Construction Method Project Example
(Takashima Apartment Building)

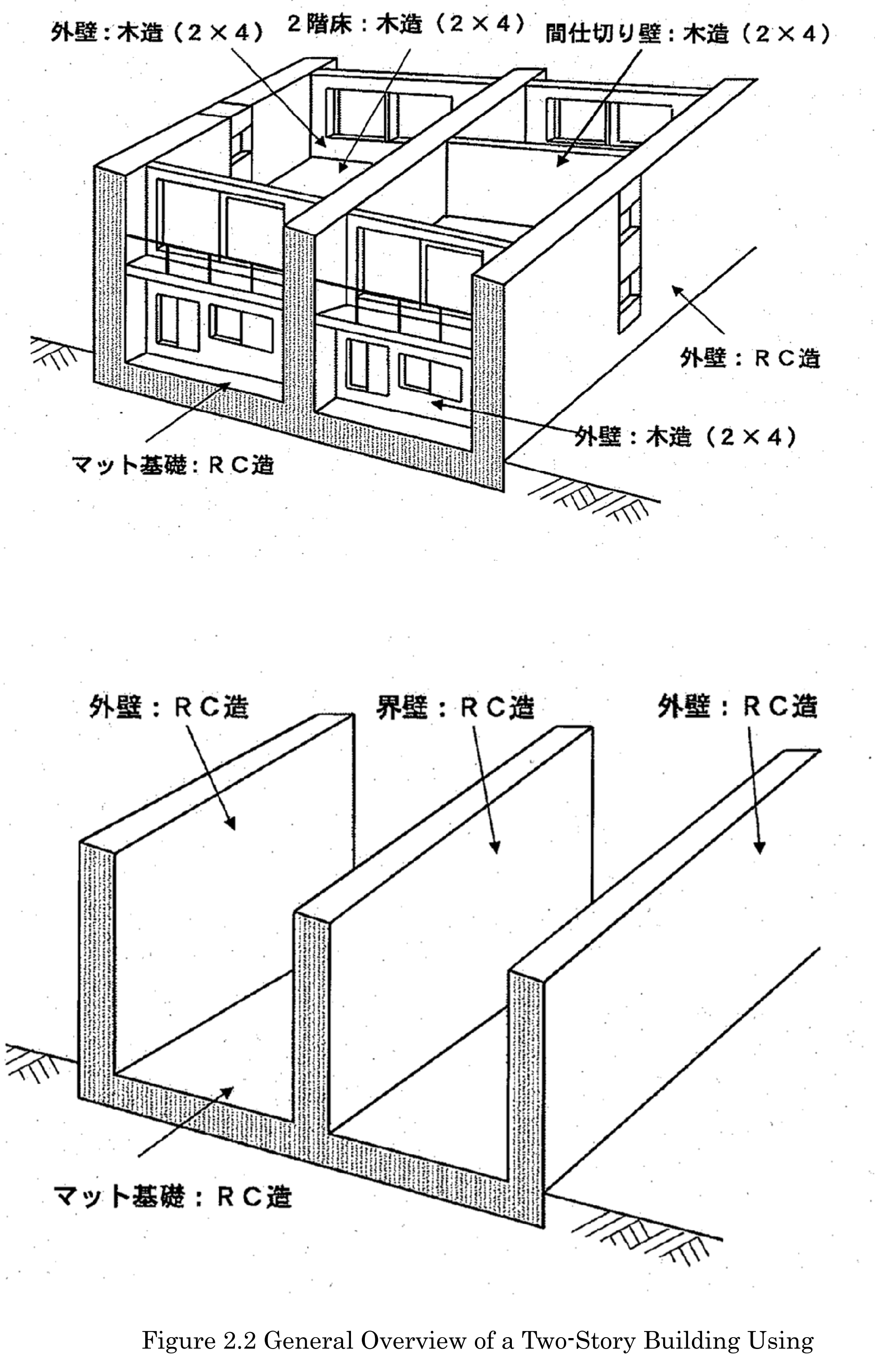


Figure 2.2 General Overview of a Two-Story Building Using the RC Upright Construction Method

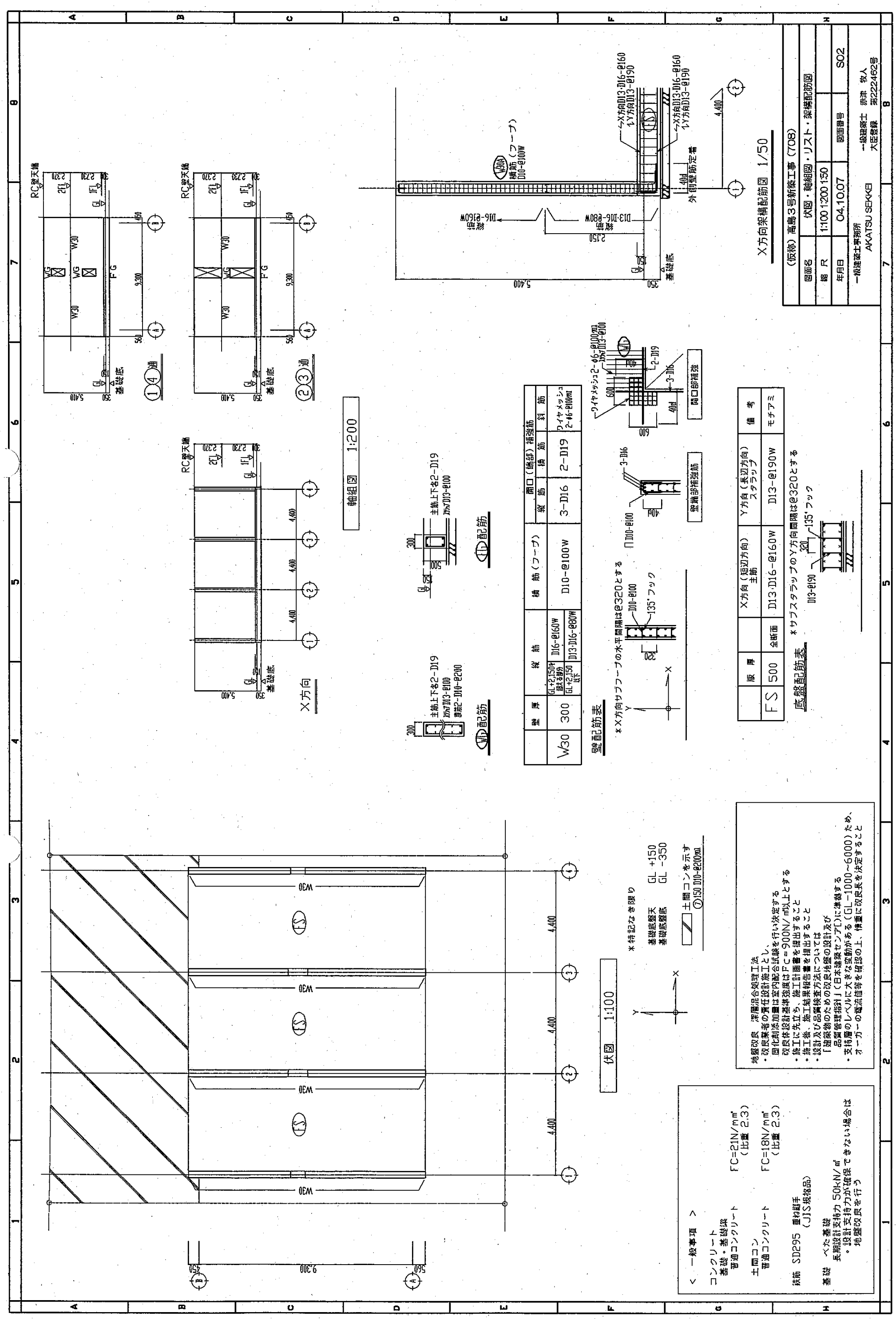


Figure 2.3 Drawings of the Subject Building for Seismic Analysis
(Two-Story RC Upright Building)

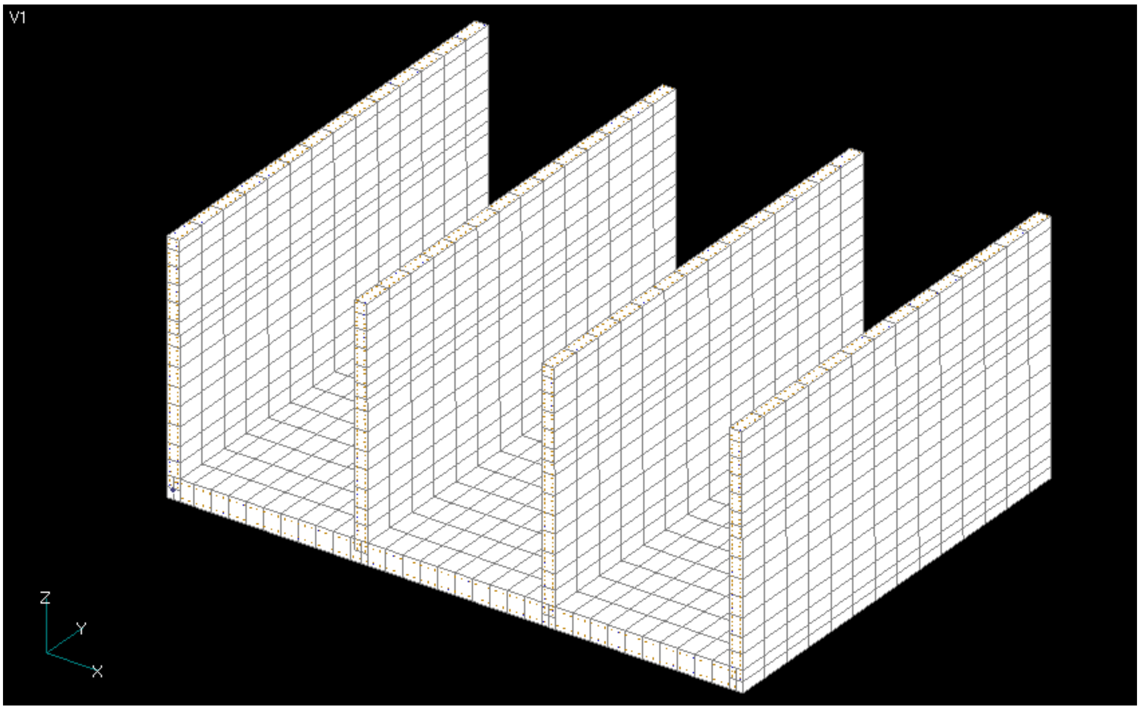


Figure 2.4 Analytical Model (Overall View with Wall and Slab Thicknesses Indicated)

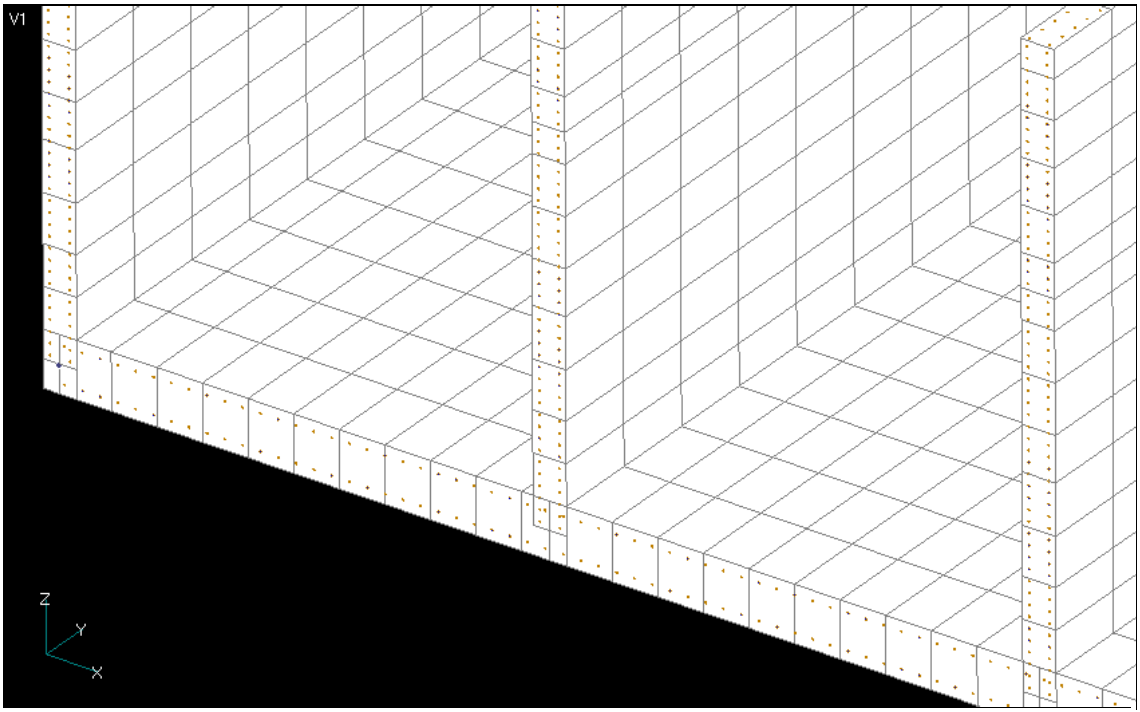


Figure 2.5 Analytical Model (Enlarged View with Wall and Slab Thicknesses Indicated)

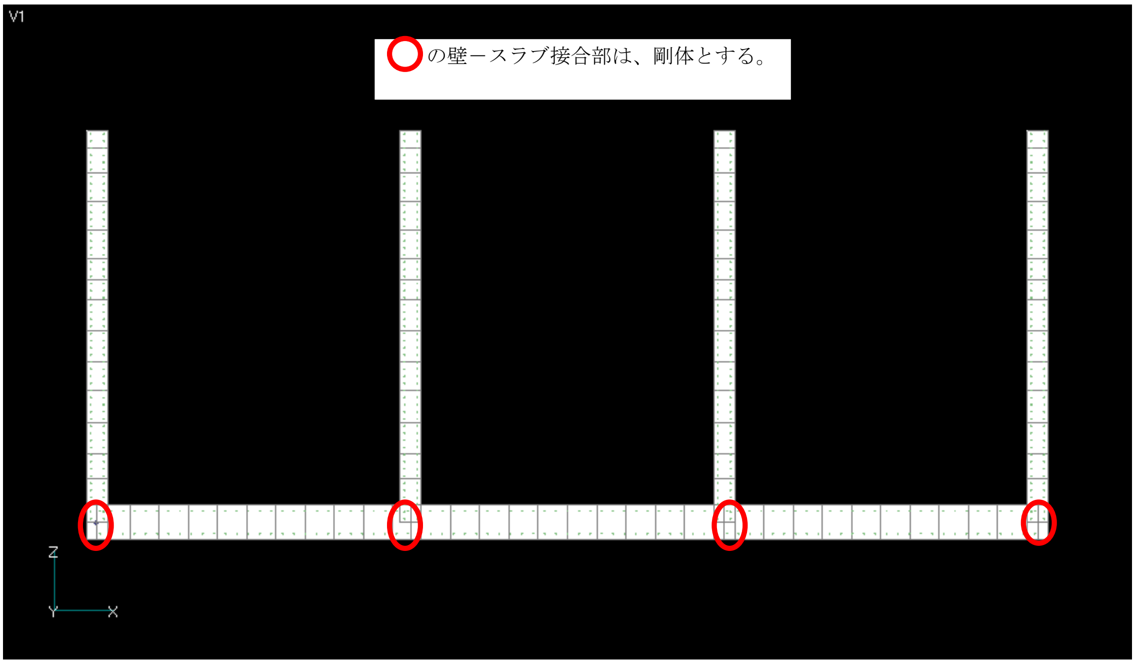


Figure 2.6 Analytical Model (Front View; Wall–Slab Connections Marked with Red Circles Are Modeled as Rigid Joints)

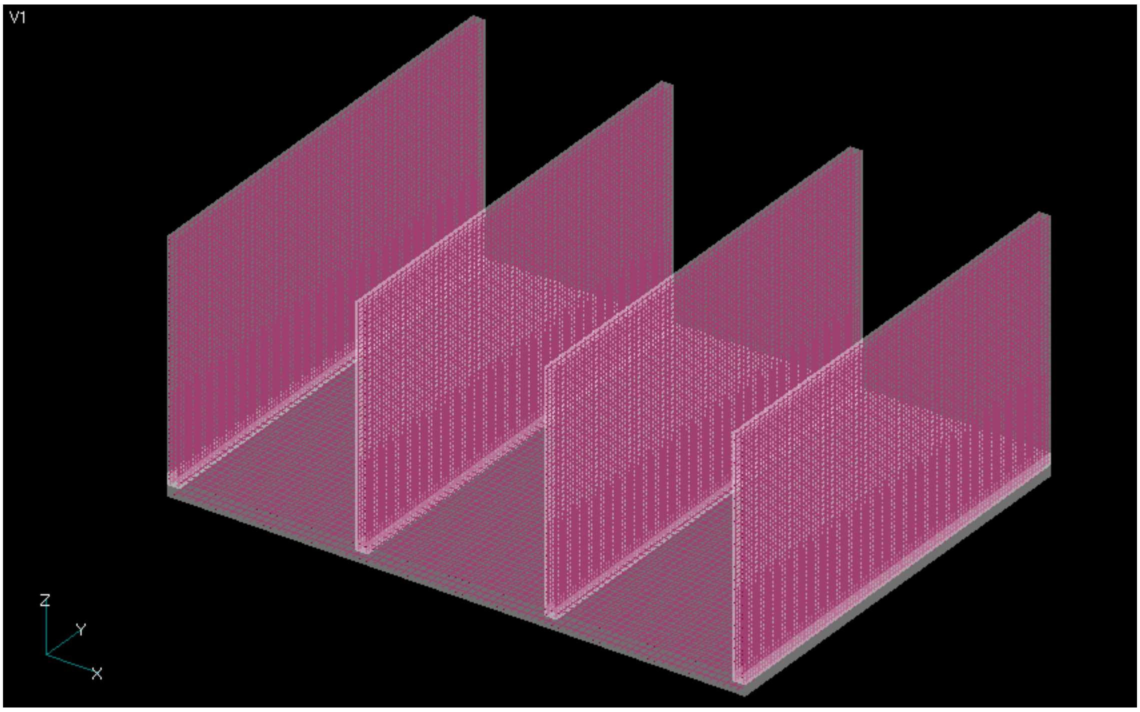


Figure 2.7 Analytical Model (Overall View with Reinforcing Bars Displayed Transparently)

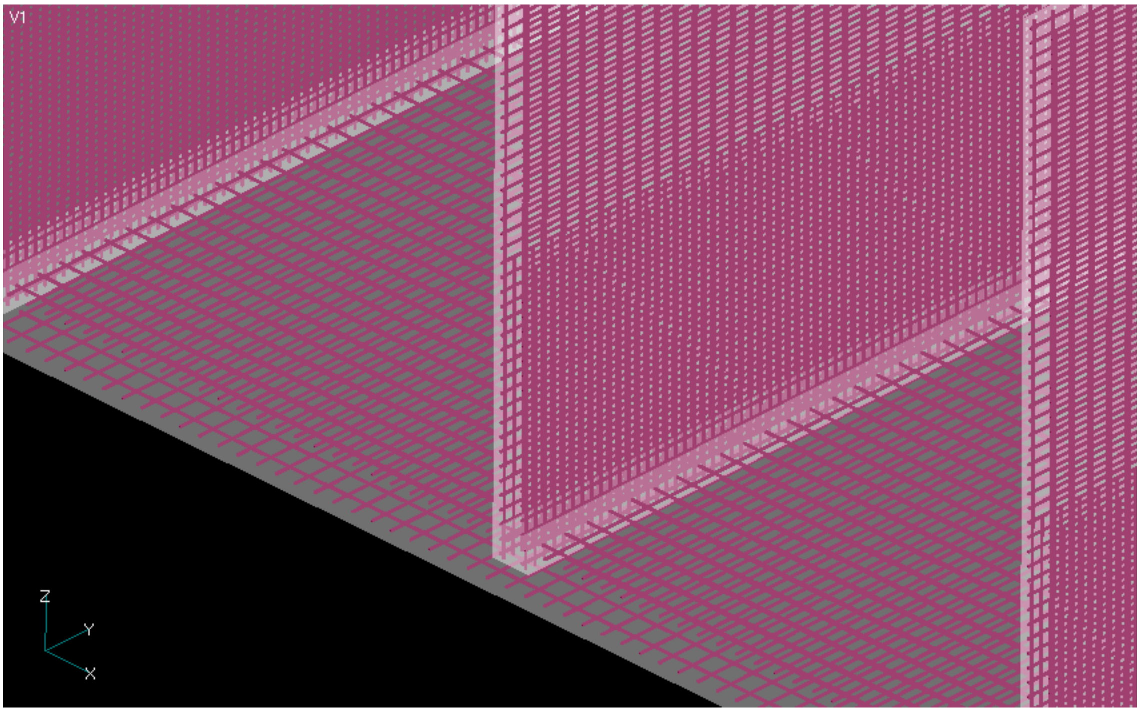


Figure 2.8 Analytical Model (Detailed View with Reinforcing Bars Displayed Transparently)

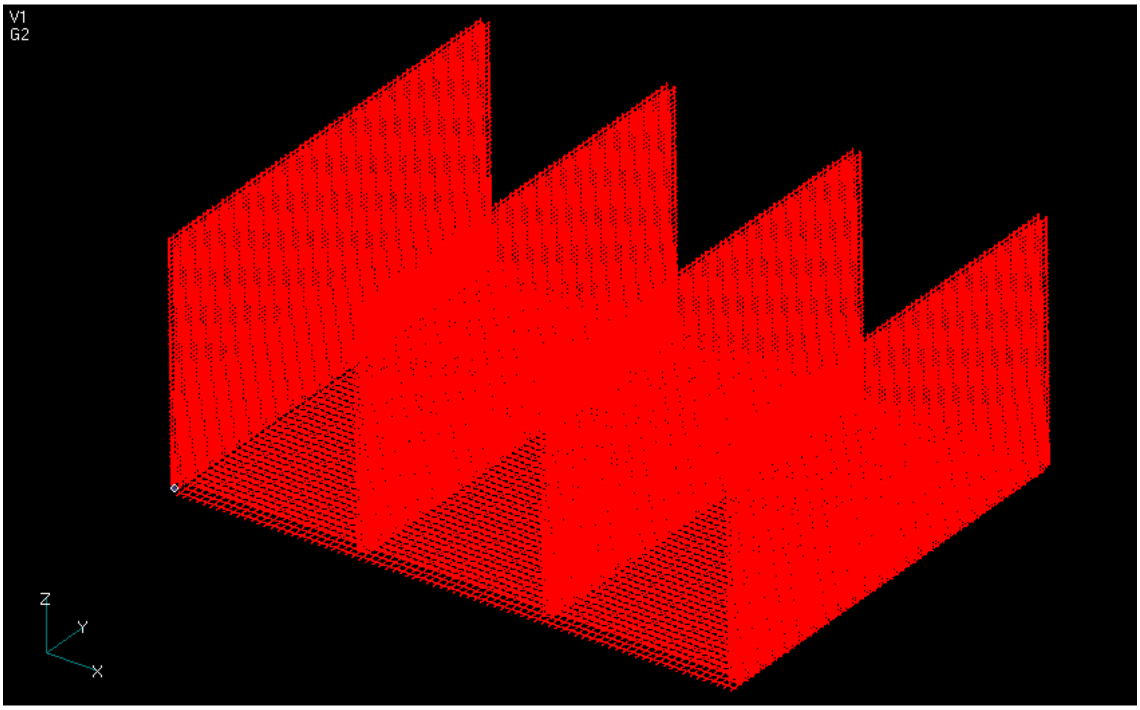


Figure 2.9 Analytical Model (Overall View Showing Reinforcing Bars Only)

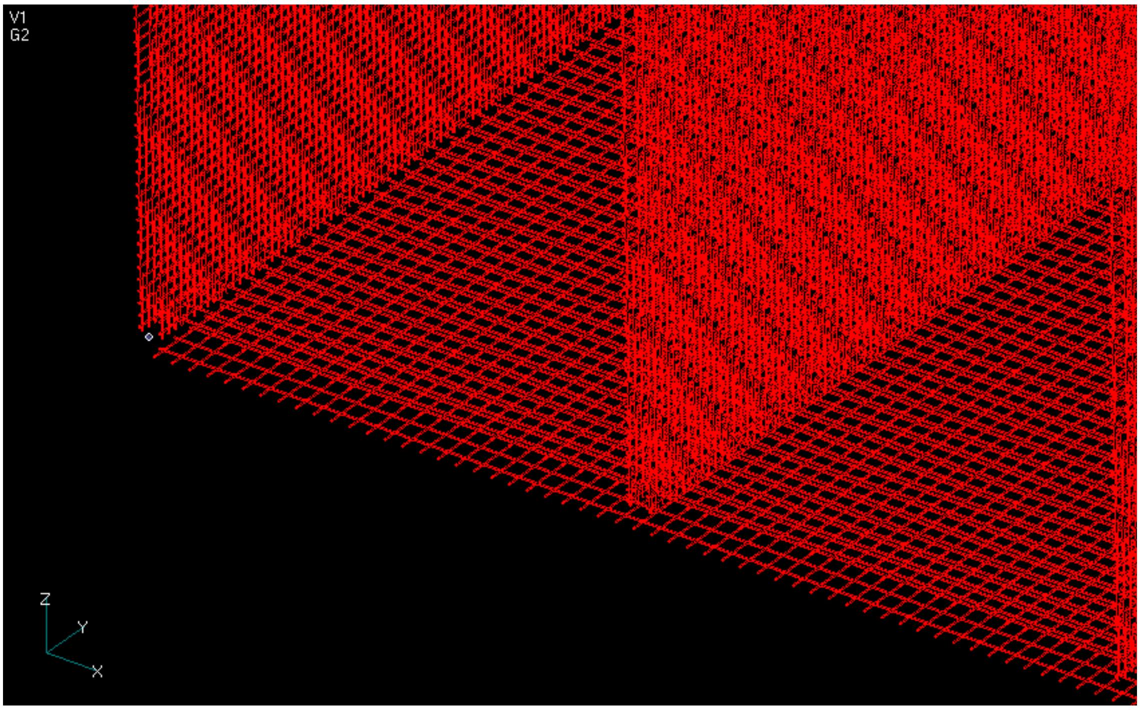


Figure 2.10 Analytical Model (Enlarged View Showing Reinforcing Bars Only)

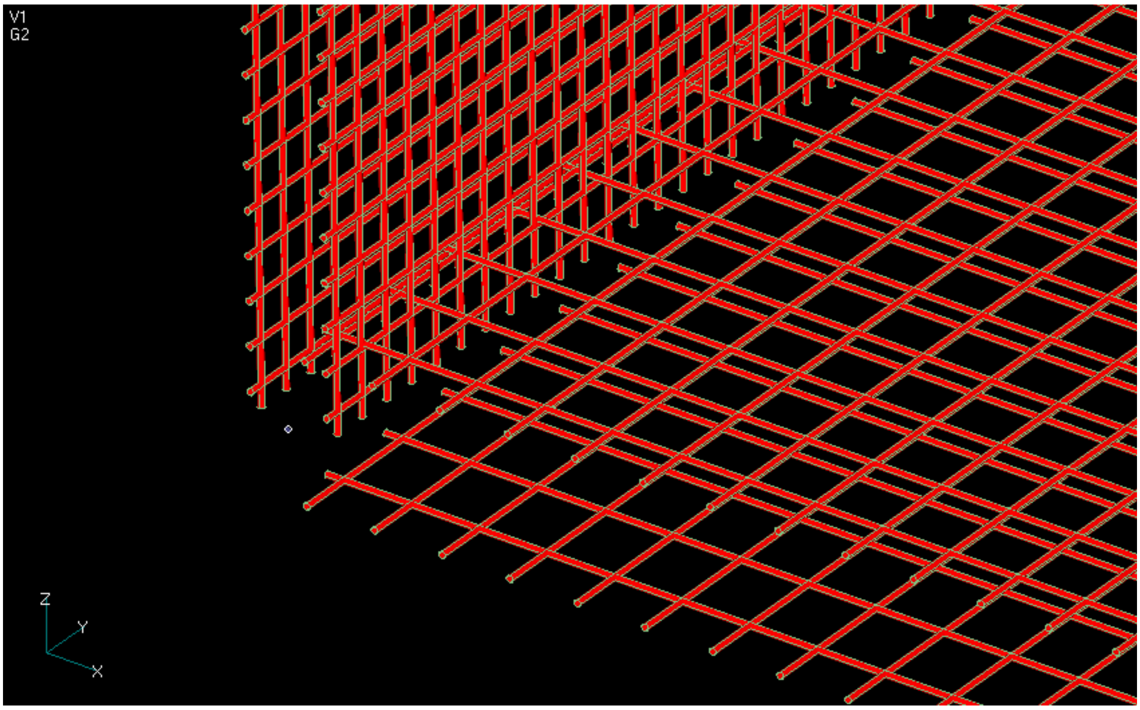


Figure 2.11 Analytical Model (Enlarged View Showing Reinforcing Bars Only)

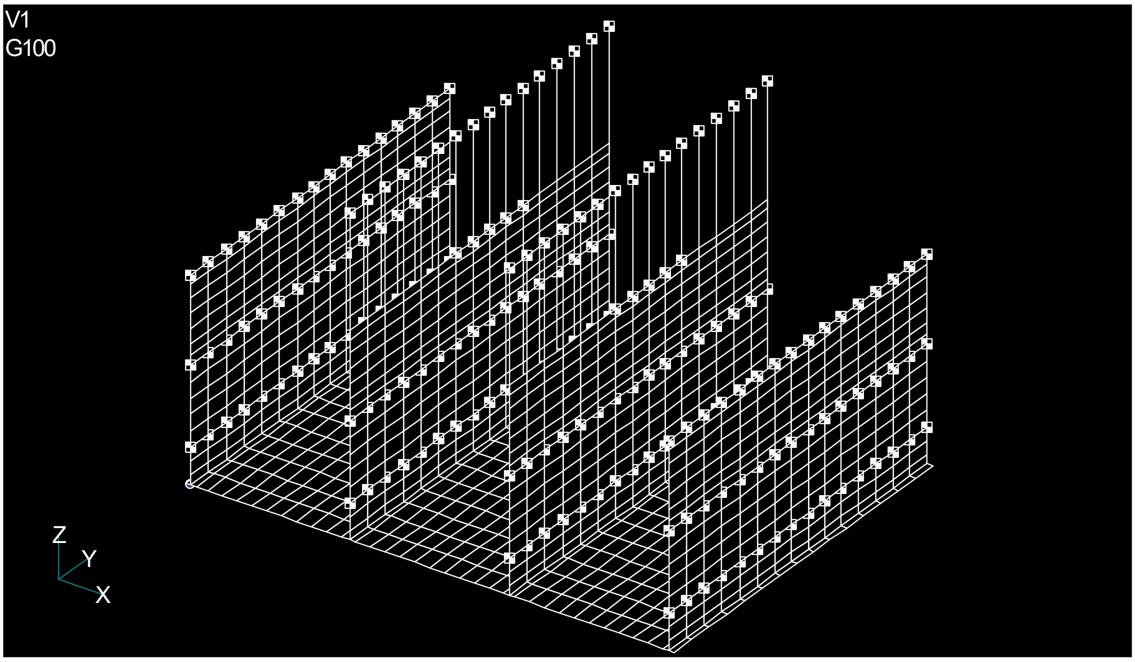


Figure 2.12 Analytical Model (Overall View Showing the Shell Element Model and Applied Additional Masses)

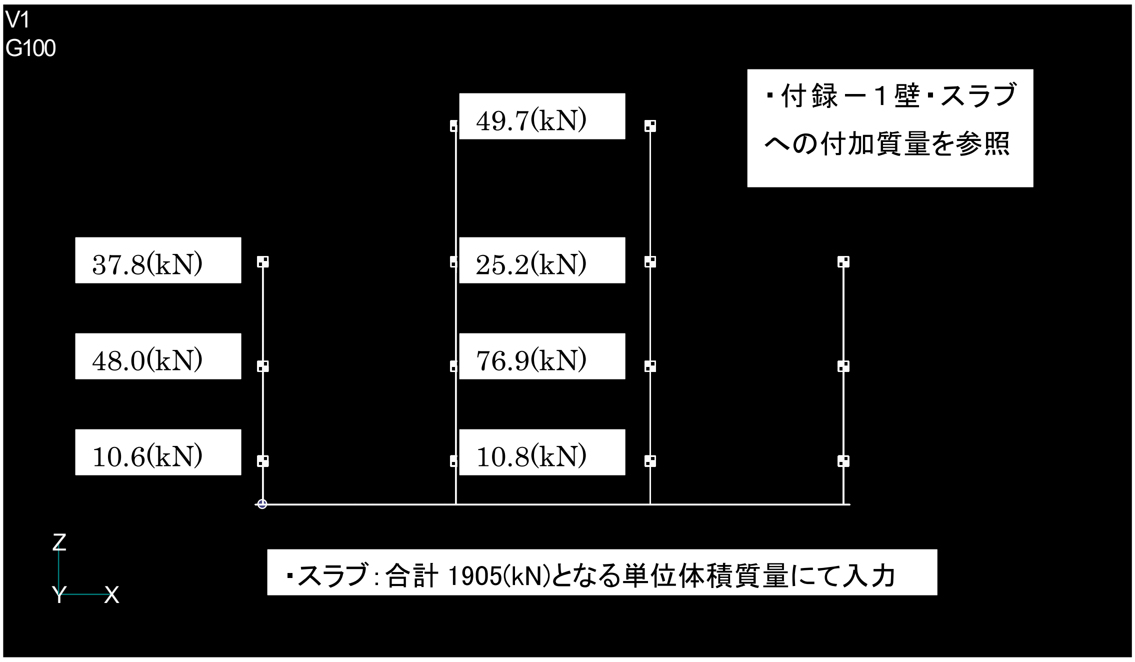


Figure 2.13 Analytical Model (Front View Showing the Shell Element Model and Applied Additional Masses)

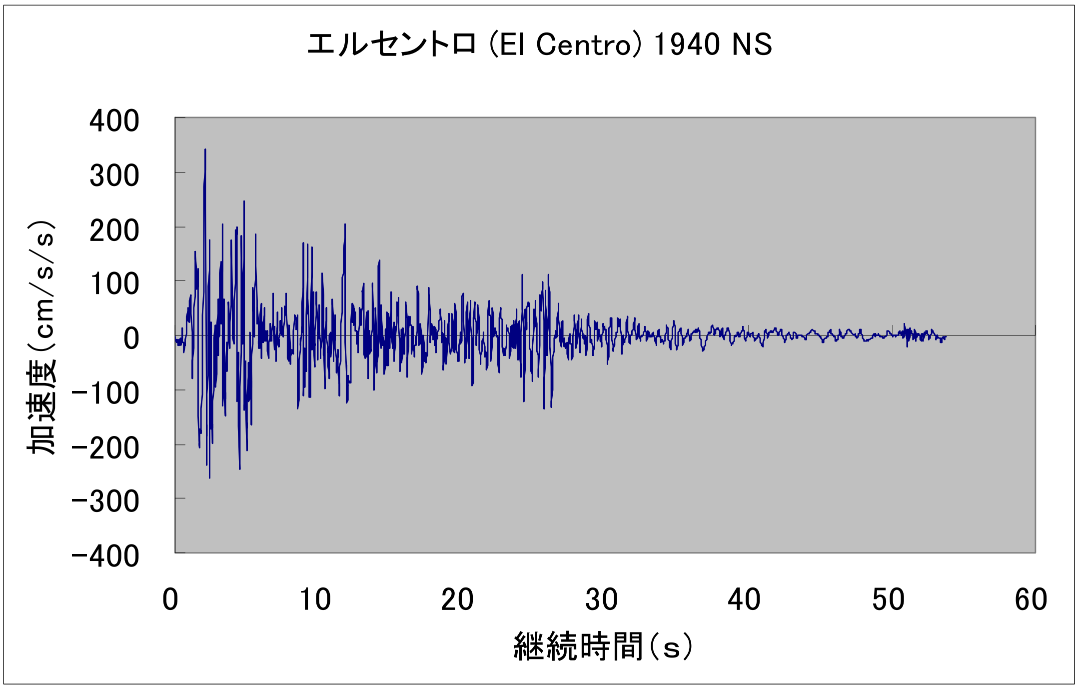


Figure 2.14 El Centro NS Ground Motion Record (Peak Ground Acceleration = 341.70 cm/s²)

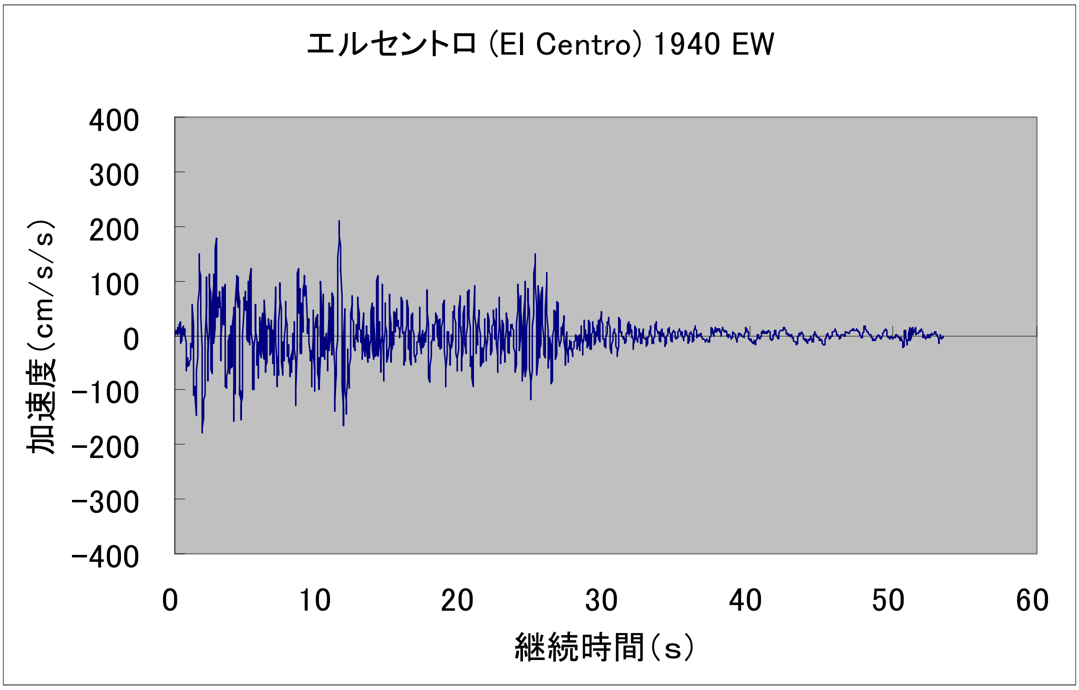


Figure 2.15 El Centro EW Ground Motion Record (Peak Ground Acceleration = 210.10 cm/s²)

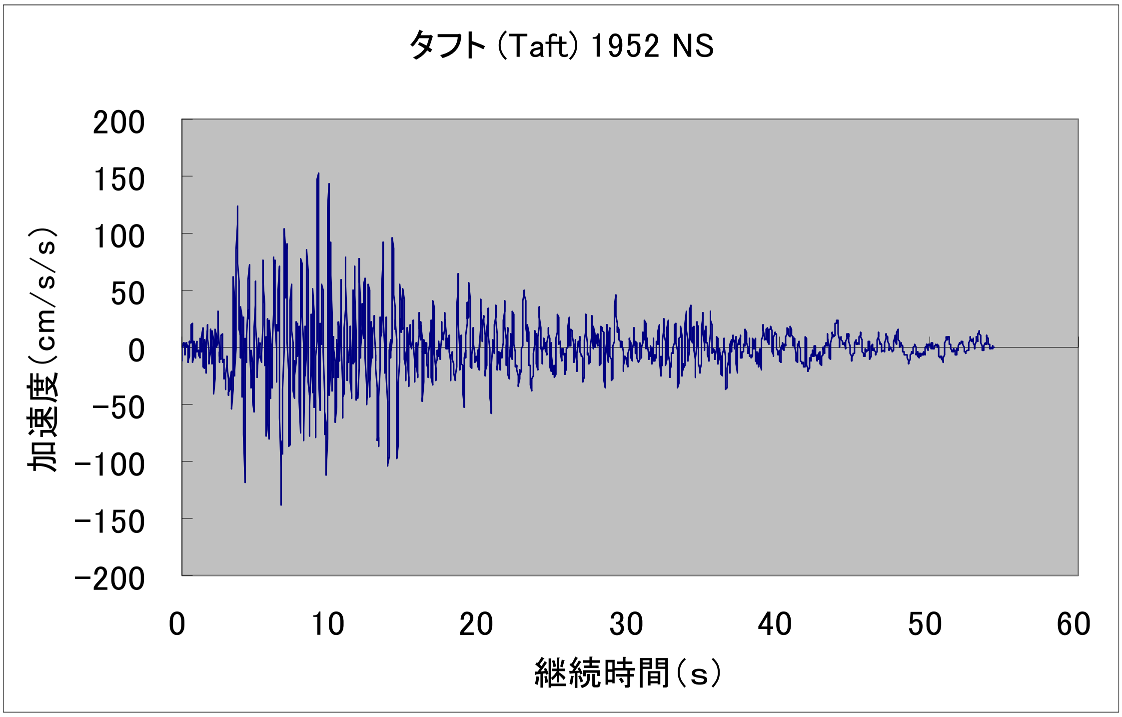


Figure 2.16 Taft NS Ground Motion Record (Peak Ground Acceleration = 152.70 cm/s²)

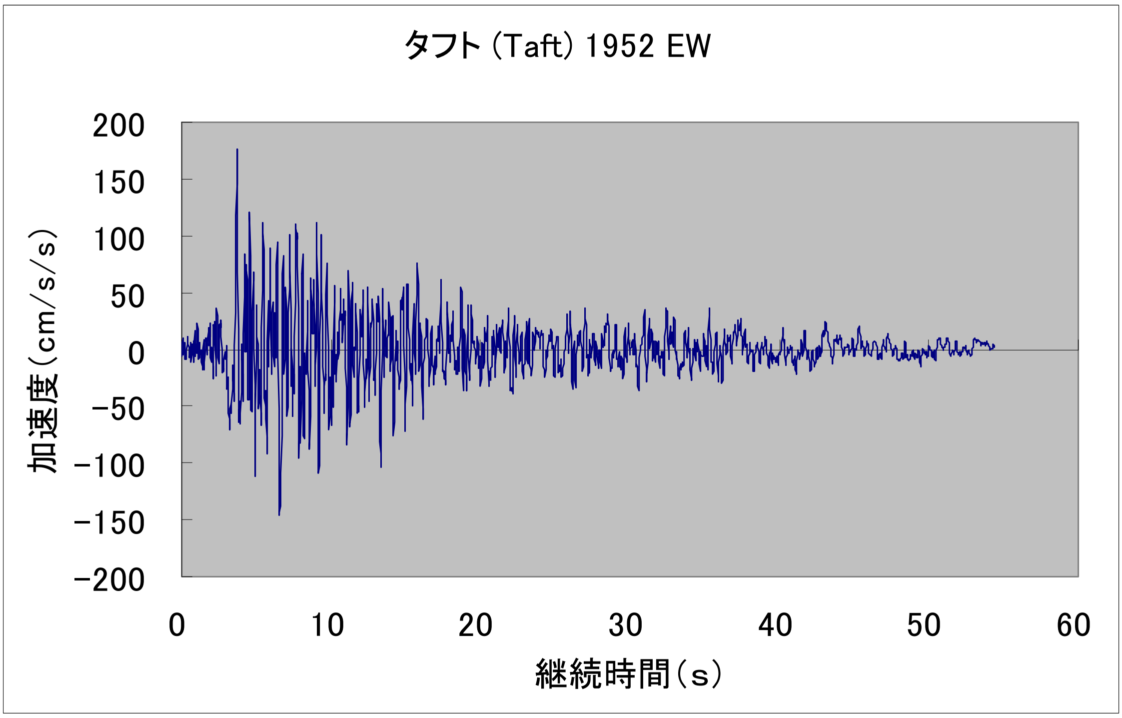


Figure 2.17 Taft EW Ground Motion Record (Peak Ground Acceleration = 175.90 cm/s²)

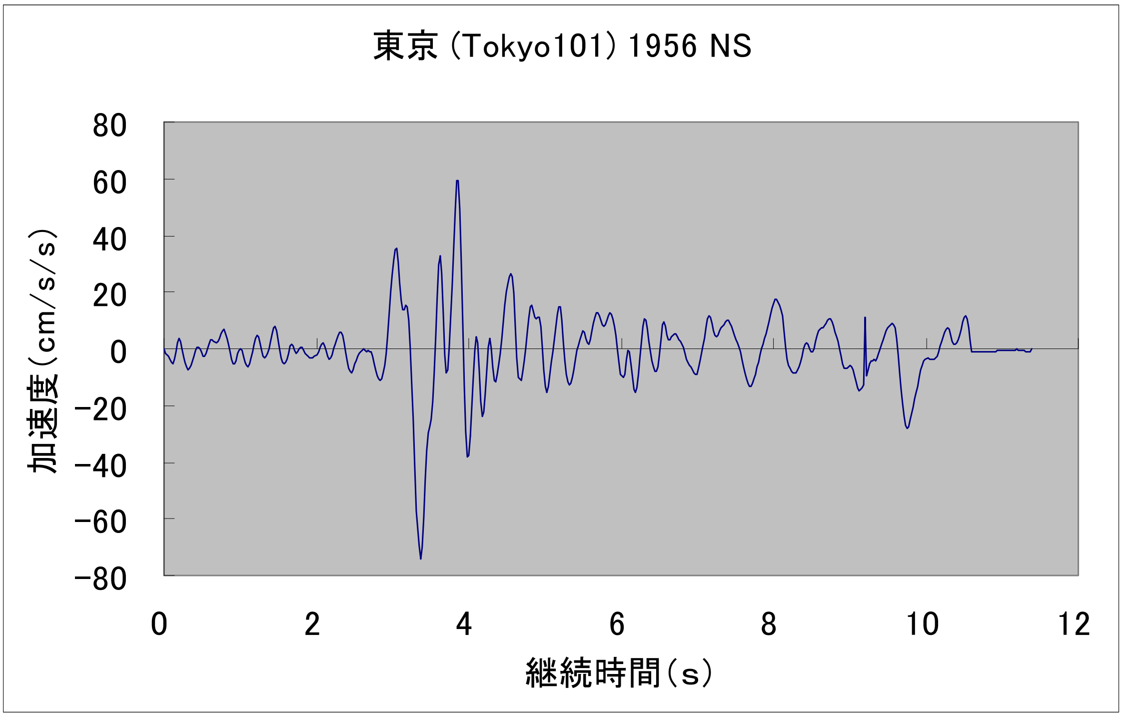


Figure 2.18 Tokyo NS Ground Motion Record (Peak Ground Acceleration = 74.00 cm/s^2)

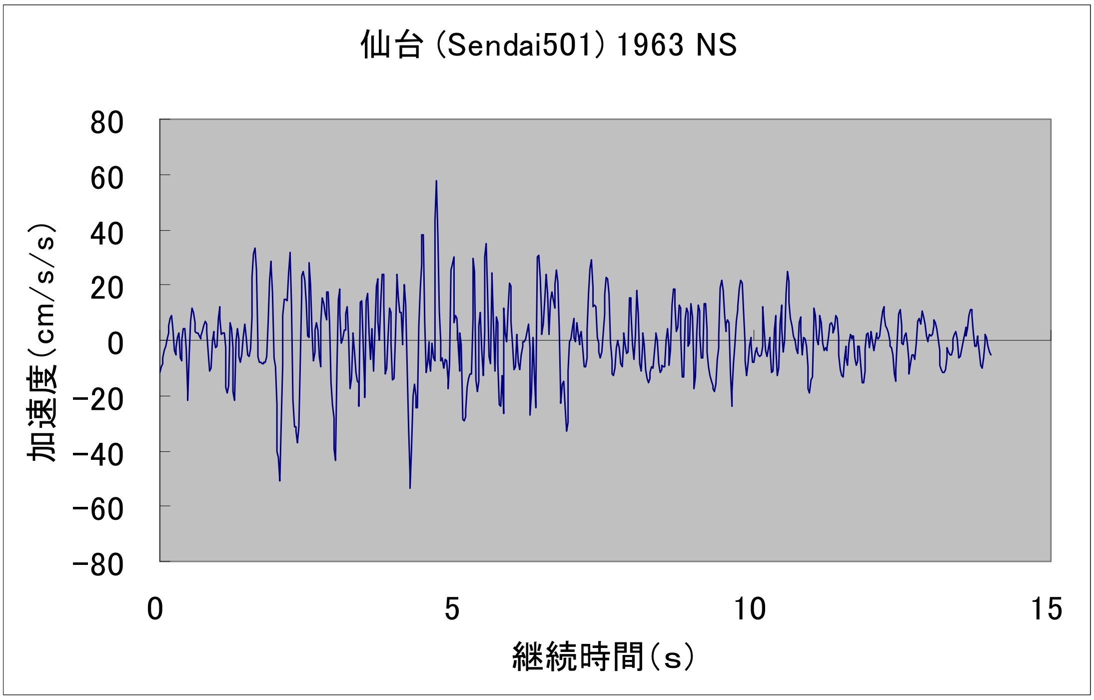


Figure 2.19 Sendai NS Ground Motion Record (Peak Ground Acceleration = 57.50 cm/s²)

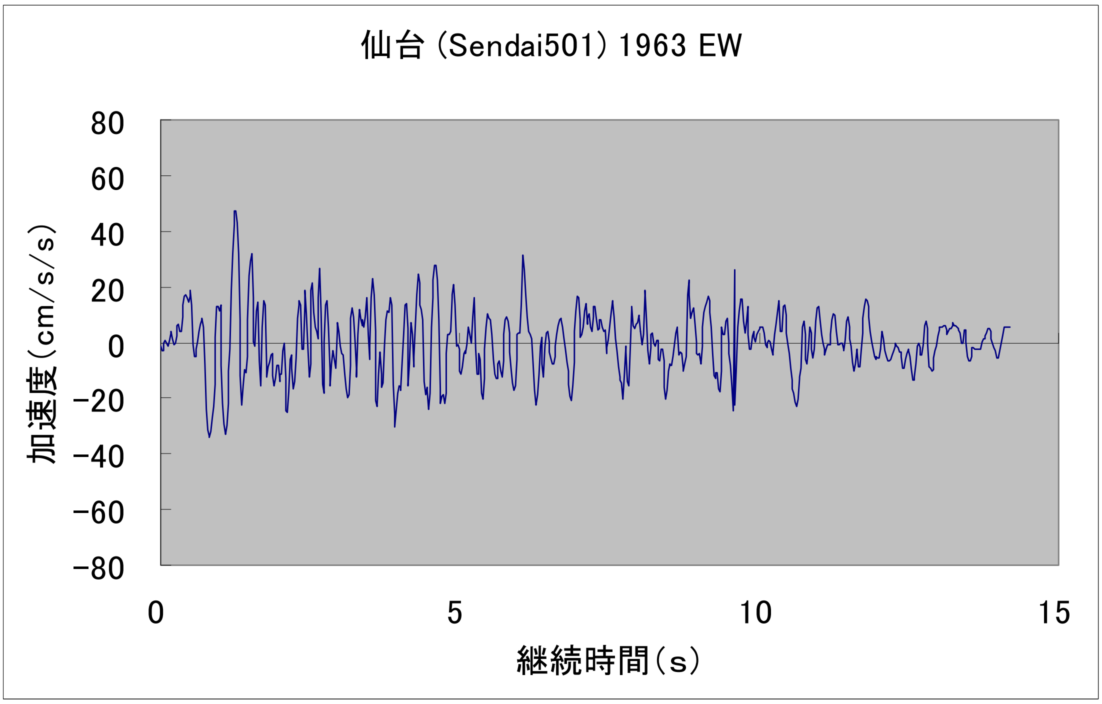


Figure 2.20 Sendai EW Ground Motion Record (Peak Ground Acceleration = 47.50 cm/s²)

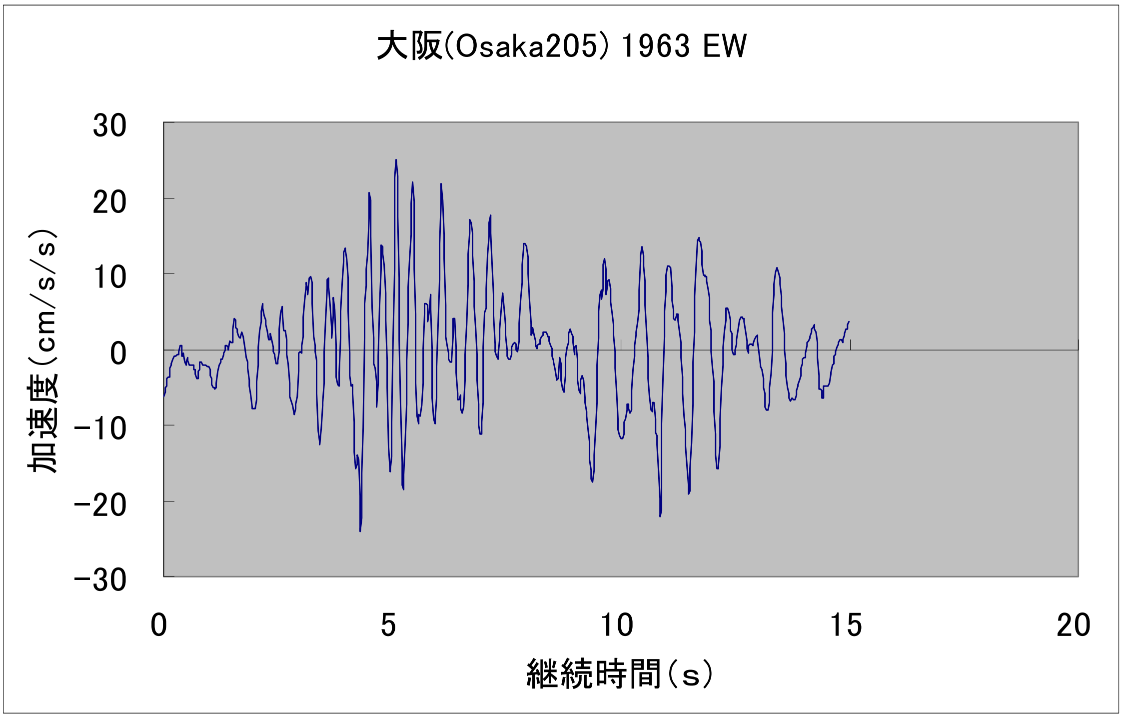


Figure 2.21 Osaka EW Ground Motion Record (Peak Ground Acceleration = 25.00 cm/s²)

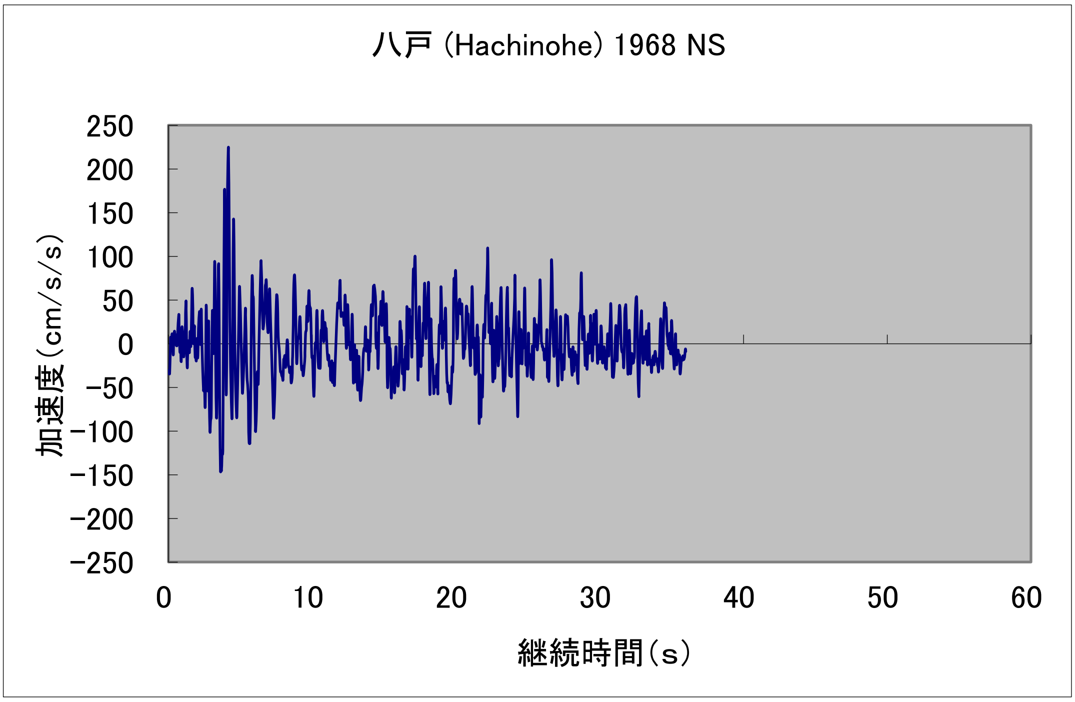


Figure 2.22 Hachinohe NS Ground Motion Record (Peak Ground Acceleration = 225.00 cm/s²)

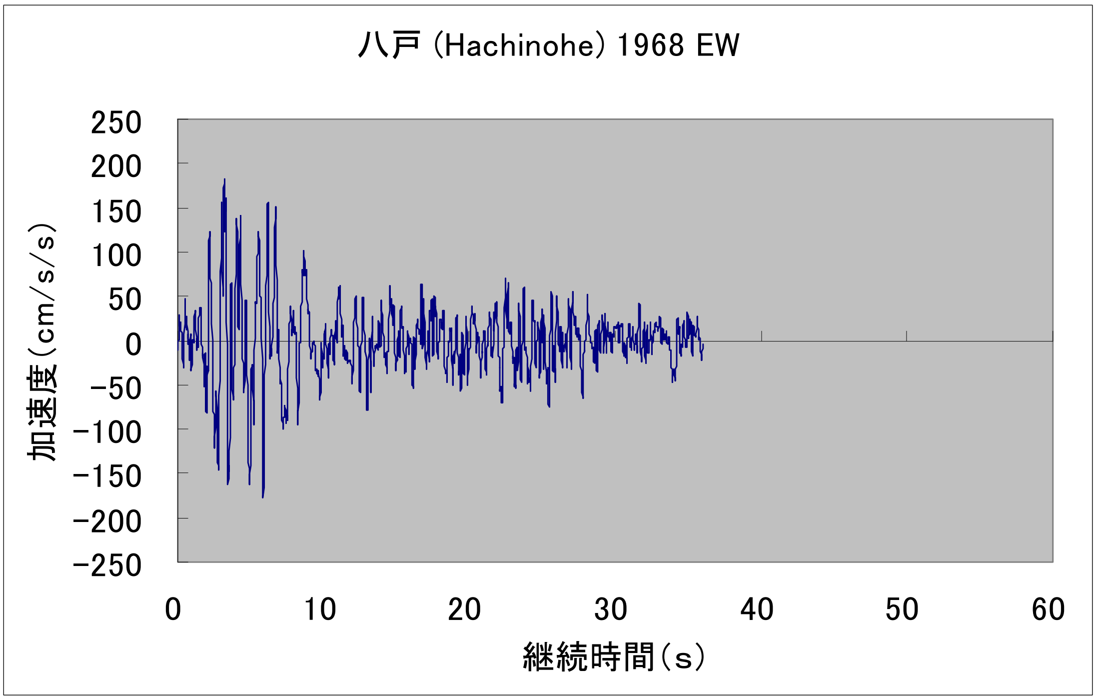


Figure 2.23 Hachinohe EW Ground Motion Record (Peak Ground Acceleration = 182.90 cm/s²)

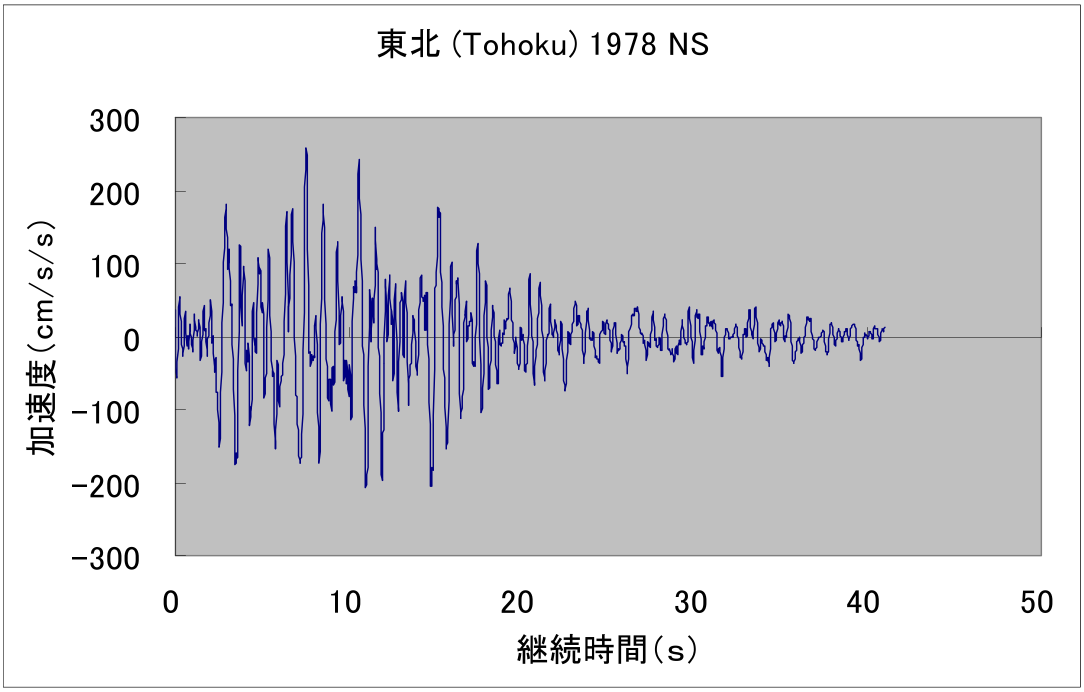


Figure 2.24 Tohoku NS Ground Motion Record (Peak Ground Acceleration = 258.20 cm/s^2)

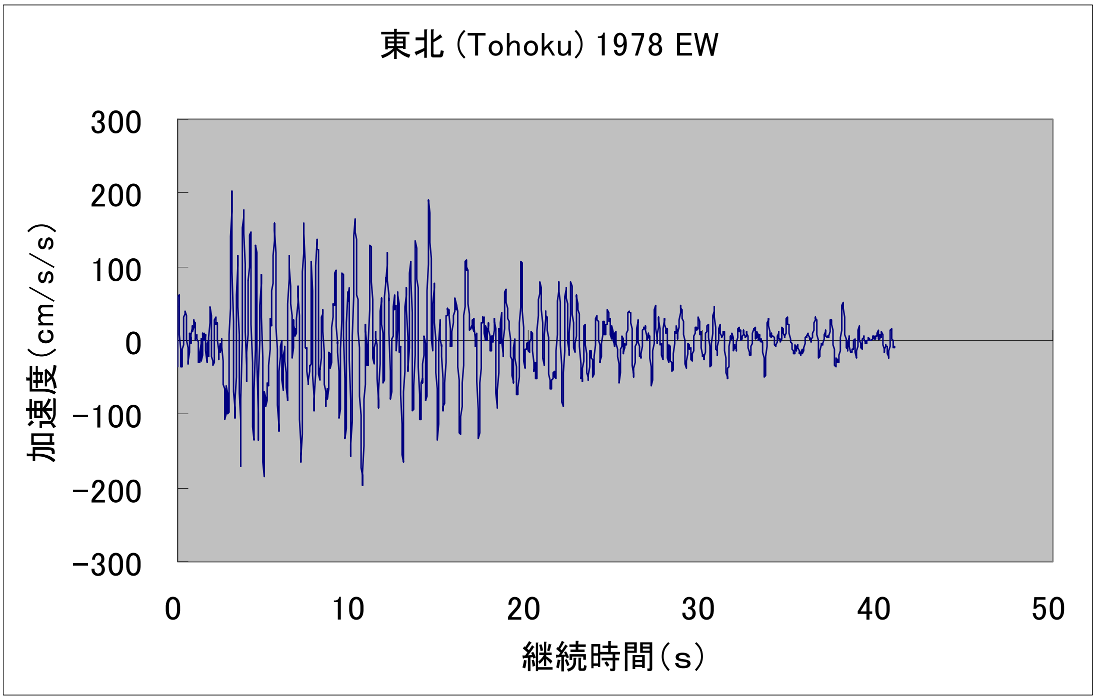


Figure 2.25 Tohoku EW Ground Motion Record (Peak Ground Acceleration = 202.50 cm/s^2)

3. Analysis Results and Discussion

3.1 Eigenvalue Analysis Results

Table 3.1 presents the eigenvalue analysis results for the two-story building constructed using the RC Upright Construction Method, including the first through twelfth vibration modes. The results include the natural frequencies, participation factors in the three translational directions, and the characteristics of each vibration mode.

In simple terms, the **natural frequency** represents the number of vibration cycles that occur in one second. It varies depending on the structural configuration, stiffness, and mass of a building or other physical object.

The **participation factor** is an indicator that represents the contribution of each vibration mode to the overall dynamic response of the structure. In other words, it shows how strongly a particular vibration mode participates in the motion of the building.

It should be noted that the mode shape diagrams for each natural vibration mode do not represent the actual magnitude of deformation. Rather, they illustrate only the relative characteristics and patterns of vibration associated with each mode.

Incidentally, when a building or structure is subjected to vibrations having a frequency that matches one of its natural frequencies, the vibration response can be significantly amplified, resulting in large oscillations. This phenomenon is known as **resonance**.

Figures 3.1 and 3.2 show the first and second natural vibration modes of the two-story building constructed using the RC Upright Construction Method. The first and second modes correspond to the first bending vibration modes of the partition walls. The first and second natural frequencies are $f_1 = f_2 = 3.126$ Hz.

Figures 3.3 and 3.4 show the third and fourth natural vibration modes of the building. These modes correspond to the first bending vibration modes of the exterior walls, and the third and fourth natural frequencies are $f_3 = f_4 = 4.678$ Hz.

Figure 3.5 shows the fifth natural vibration mode of the building. This mode corresponds to the first torsional vibration mode of the partition walls, with a natural frequency of f_5 = 5.146 Hz. In this mode, the two partition walls undergo torsional vibration in phase with each other.

Figure 3.6 shows the sixth natural vibration mode of the building. This mode corresponds to the second torsional vibration mode of the partition walls, with a natural frequency of f_6 = 5.146 Hz. In this mode, the two partition walls undergo torsional vibration out of phase with each other.

Figures 3.7 and 3.8 show the seventh and eighth natural vibration modes of the building. These modes correspond to the first torsional vibration modes of the exterior walls, and the seventh and eighth natural frequencies are $f_7 = f_8$ = 7.262 Hz.

Focusing on the **participation factors** listed in **Table 3.1**, it can be seen that the participation factors in the **X-direction** (the weak-axis direction of the partition walls and exterior walls) are relatively large for the first through fourth vibration modes. This indicates that **bending vibrations in the weak-axis direction of the partition walls and exterior walls are more likely to occur**, and these modes contribute significantly to the dynamic response of the structure.

As a validation check for the eigenvalue analysis results of the two-story building constructed using the **RC Upright Construction Method**, the theoretical values and analytical values are compared and examined. However, this verification is conducted using a simplified model that excludes additional loads such as those from the roof.

The first natural frequency, f_1, of an unreinforced concrete exterior wall with a height of **5.4 m**, thickness of **0.3 m**, and width of **10.31 m** can be obtained using the following equation. As the boundary condition, the lower end of the exterior wall is assumed to be fixed.

$$f_1=(1.875)^2\times(\sqrt{}(E\times I\times g)/(\gamma\times A))/(2\pi L^2)$$

$$=(1.875)^2\times(\sqrt{}(2.17\times10^6\times(10.31\times0.3^3/12)\times9.8)/(2.4\times10.31\times0.3))/(2\pi 5.4^2)$$

$$\therefore\ f_1=4.95(\mathrm{Hz})$$

The first natural frequency, f_1, of the RC Upright Construction Method model without reinforcement, which was analyzed separately under the boundary condition that the bottom surface of the slab was fixed, was **f_1 = 4.9 Hz** for the first bending vibration mode of the exterior wall.

This result is consistent with the theoretical value described above. Therefore, the analytical model of the two-story building constructed using the **RC Upright Construction Method** is considered to be valid.

In addition, according to a separate measurement survey conducted on the subject building, a vibration generator was installed on the second floor to measure the natural frequency. As a result, the first natural frequency of the building was found to be f_1 = 6.2 Hz (T_1 = 0.16 s), indicating that the building has a relatively stiff structural system.

The first natural frequency obtained from the present eigenvalue analysis was f_1 = 3.13 Hz (T_1 = 0.32 s), which is lower than the measured value. This difference is considered to be due to the fact that, in this analysis, the stiffness of the roof, wooden walls, and floors was neglected. In contrast, the actual building shown in Figure 2.1, constructed using the RC Upright Construction Method, includes a roof as well as wooden walls and floors. These elements are considered to increase the overall stiffness of the building, resulting in a higher first natural frequency.

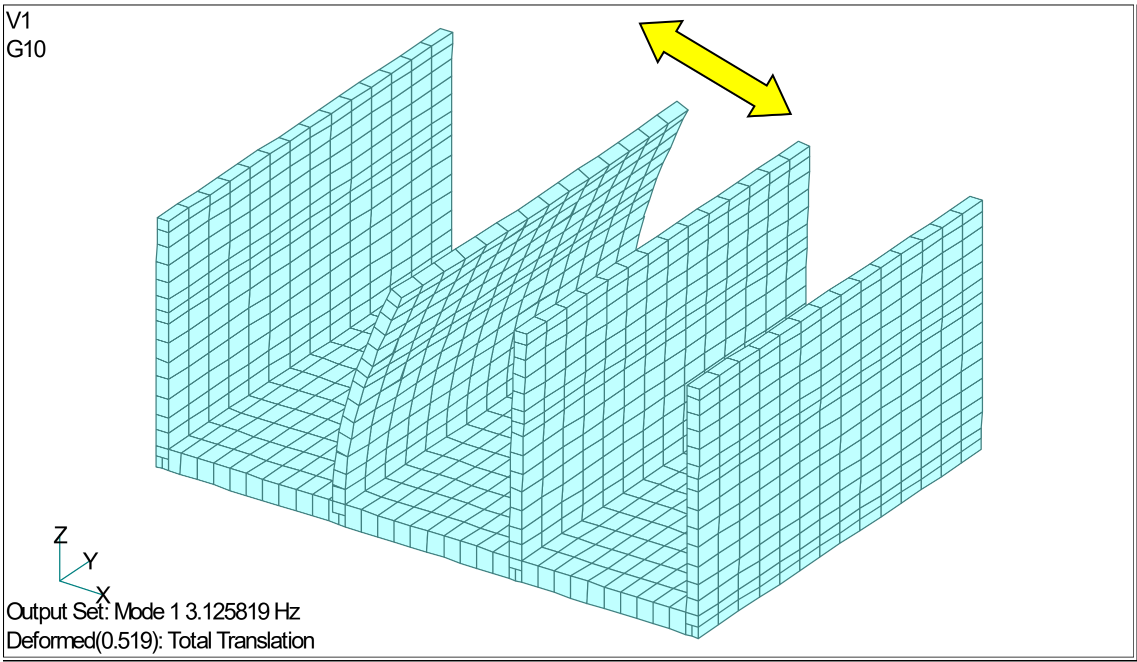


Figure 3.1 First Natural Vibration Mode (First Bending Mode of the Partition Wall, f_1 = 3.126 Hz)

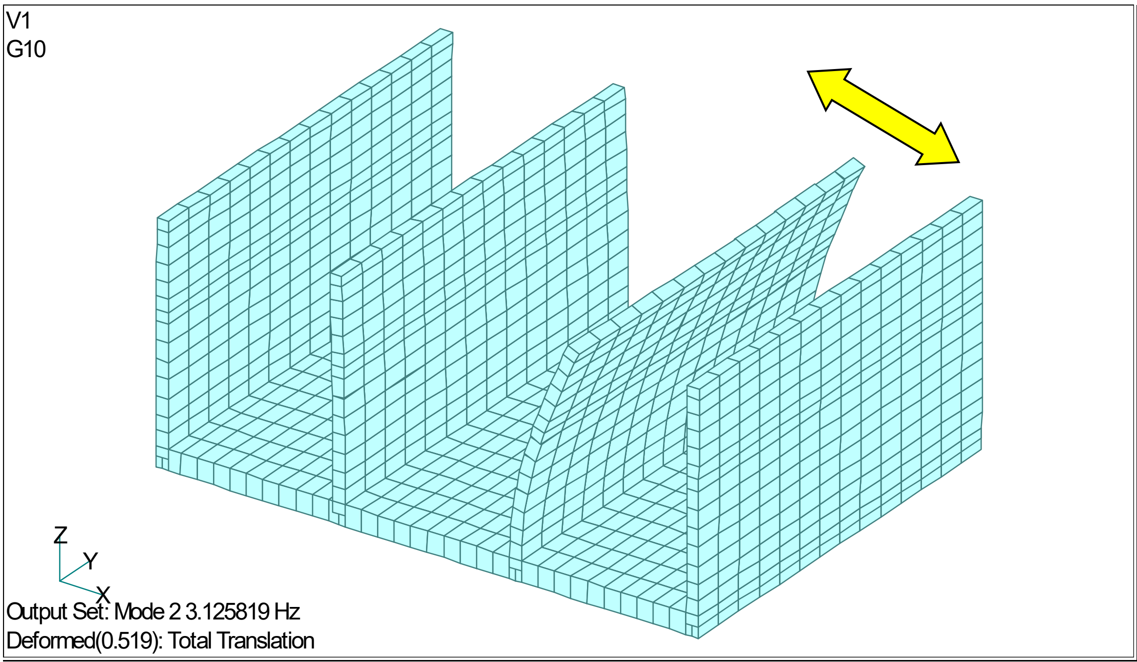


Figure 3.2 Second Natural Vibration Mode (First Bending Mode of the Partition Wall, f_2 = 3.126 Hz)

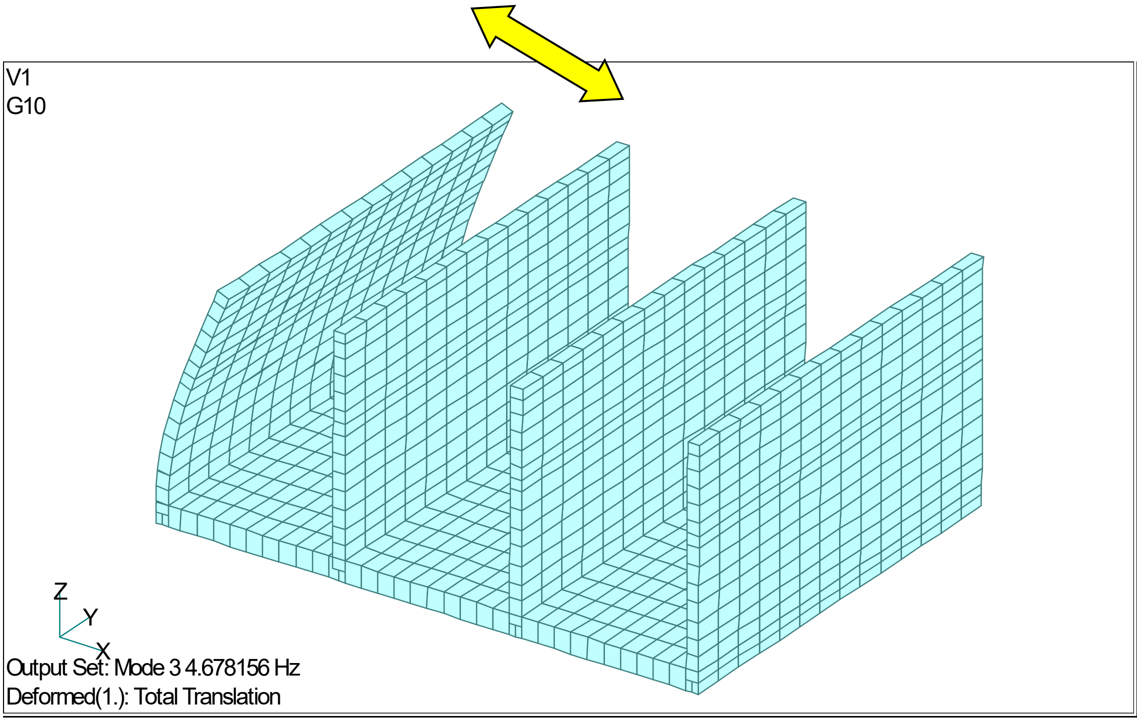


Figure 3.3 Third Natural Vibration Mode (First Bending Mode of the Exterior Wall, f_3 = 4.678 Hz)

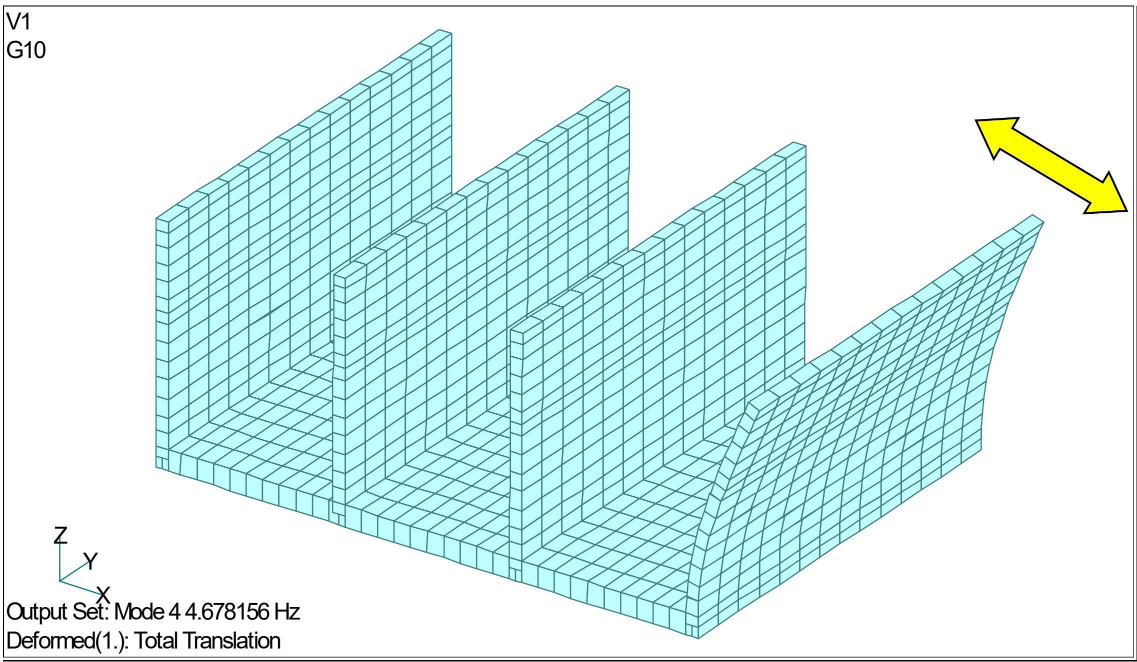


Figure 3.4 Fourth Natural Vibration Mode (First Bending Mode of the Exterior Wall, f_4 = 4.678 Hz)

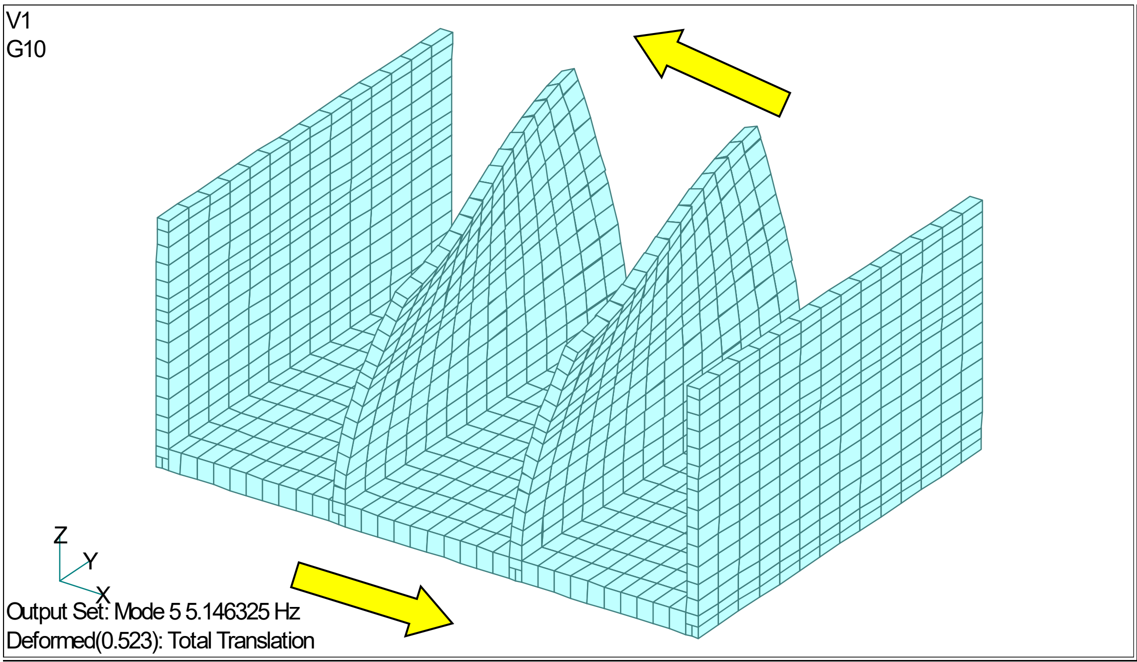


Figure 3.5 Fifth Natural Vibration Mode (First Torsional Mode of the Partition Walls, f_5 = 5.146 Hz; the Two Partition Walls Undergo Torsional Vibration in Phase)

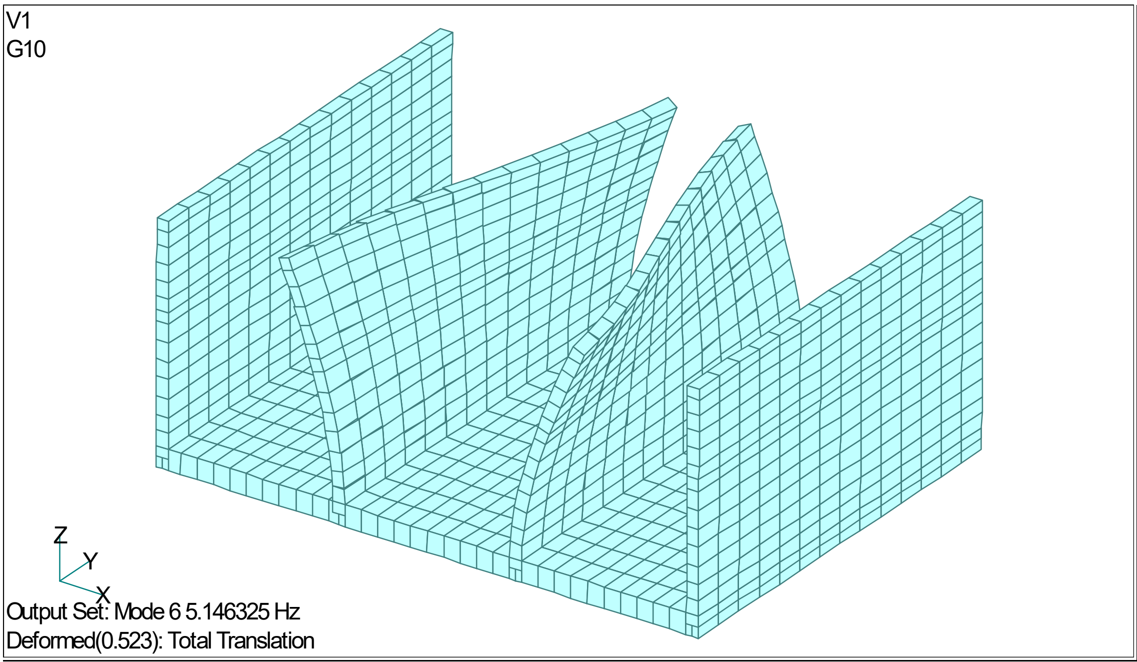


Figure 3.6 Sixth Natural Vibration Mode (Second Torsional Mode of the Partition Walls, f_6 = 5.146 Hz; the Two Partition Walls Undergo Torsional Vibration Out of Phase)

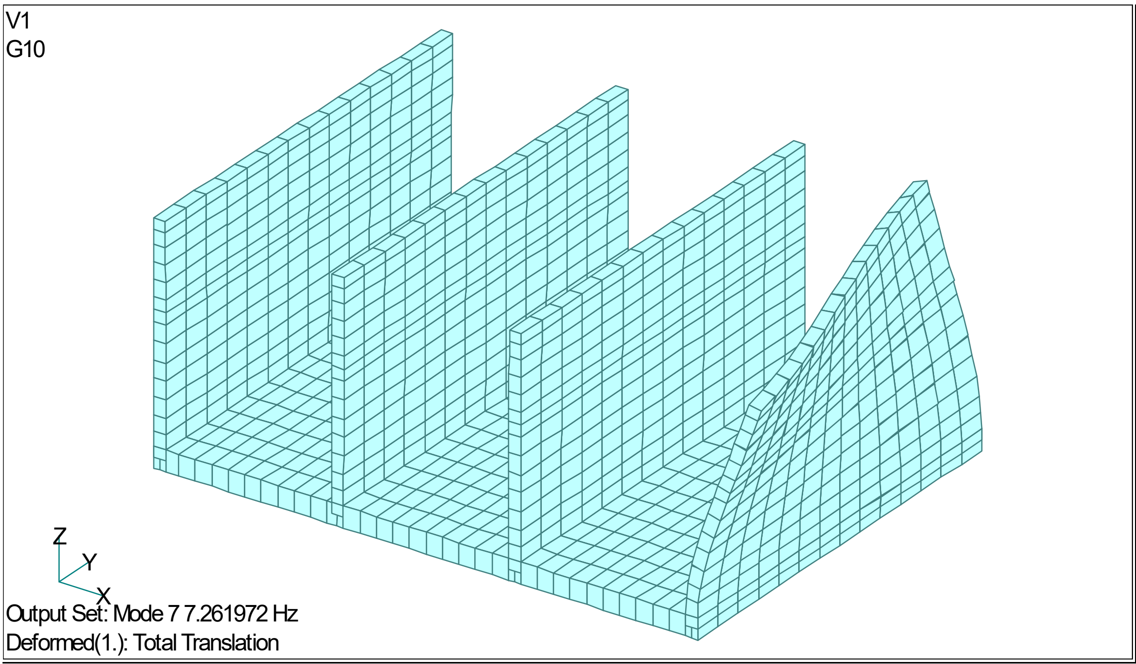


Figure 3.7 Seventh Natural Vibration Mode (First Torsional Mode of the Exterior Walls, f_7 = 7.262 Hz)

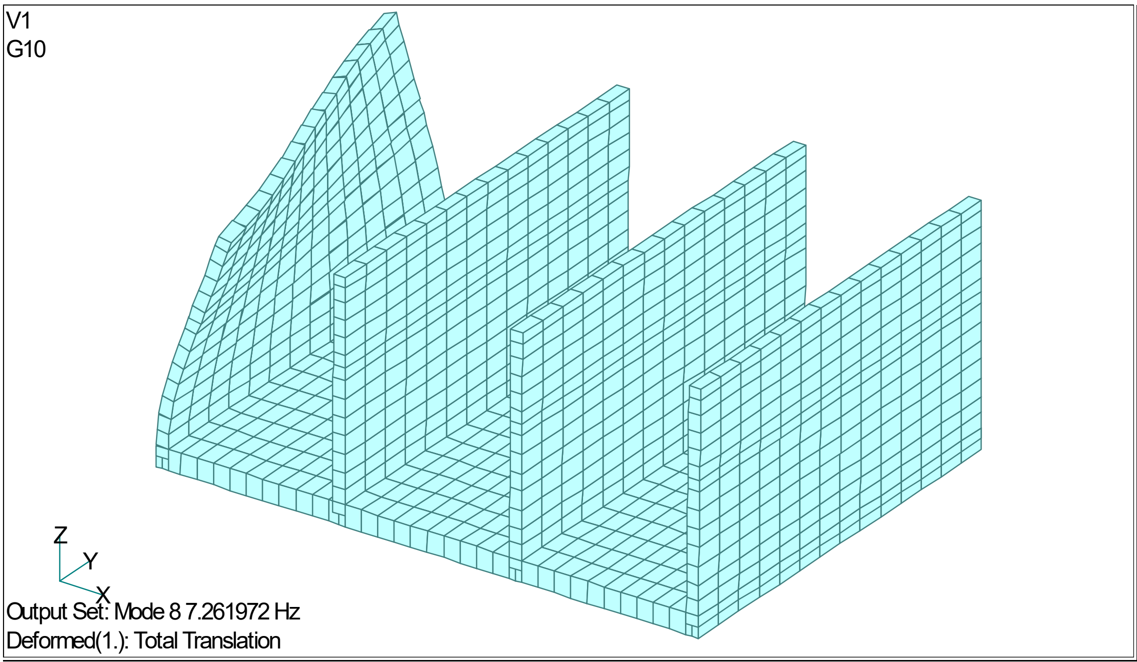


Figure 3.8 Eighth Natural Vibration Mode (First Torsional Mode of the Exterior Walls, f_8 = 7.262 Hz)

表 3.1　固有値解析結果

Table 3.1 Eigenvalue Analysis Results

Mode	Natural Frequency (Hz)	Participation Factor X	Participation Factor Y	Participation Factor Z	Vibration Mode Characteristics
1	3.1258	1.9825	−1.68E−17	−7.48E−18	First Bending Mode of Partition Wall
2	3.1258	1.7204	−1.05E−16	2.04E−17	First Bending Mode of Partition Wall
3	4.6782	1.4976	−9.84E−17	1.34E−17	First Bending Mode of Exterior Wall
4	4.6782	1.4702	5.06E−17	−2.30E−17	First Bending Mode of Exterior Wall
5	5.1463	2.24E−02	−3.90E−17	2.46E−17	First Torsional Mode of Partition Walls (Same−Phase Twisting)
6	5.1463	1.23E−03	1.76E−16	2.50E−18	Second Torsional Mode of Partition Walls (Opposite−Phase Twisting)
7	7.2620	3.87E−04	1.22E−16	−5.65E−17	First Torsional Mode of Exterior Wall
8	7.2620	4.12E−04	−3.20E−17	2.60E−17	First Torsional Mode of Exterior Wall
9	9.1584	7.16E−02	−2.01E−16	−8.27E−17	Second Bending Mode of Partition Wall
10	9.1584	0.13104	−5.25E−16	2.41E−17	Second Bending Mode of Partition Wall
11	13.910	6.82E−02	−7.59E−14	2.56E−16	Second Bending Mode of Exterior Wall
12	13.910	6.67E−02	2.72E−14	−8.46E−17	Second Bending Mode of Exterior Wall

3.2 Earthquake Response Analysis Results

A **nonlinear dynamic earthquake response analysis** was performed for the two-story building constructed using the **RC Upright Construction Method**, consisting of a static self-weight analysis followed by seismic excitation using the **El Centro NS ground motion record**.

Figure 3.8 shows the time-history of the horizontal relative displacement between the top of the exterior wall and the slab. The maximum horizontal relative displacement between the exterior wall top and the slab was **12.2 mm** at **t = 2.93 s**.

Figure 3.9 shows the time-history of the horizontal acceleration at the top of the exterior wall. The maximum horizontal acceleration at the exterior wall top was **1,098 gal** at **t = 3.14 s**.

Figure 3.10 shows the time-history of the horizontal relative displacement between the top of the interior wall and the slab. The maximum horizontal relative displacement between the interior wall top and the slab was **26.4 mm** at **t = 5.02 s**.

Figure 3.11 shows the time-history of the horizontal acceleration at the top of the interior wall. The maximum horizontal acceleration at the interior wall top was **883 gal** at **t = 5.31 s**.

Comparing the displacement and acceleration time histories shown in **Figures 3.8 and 3.9** for the exterior wall with those shown in **Figures 3.10 and 3.11** for the interior wall, it can be seen that the response period of the interior wall is slightly longer. This is considered to be due to the larger roof load carried by the interior wall, which increases its effective mass.

Figure 3.12 shows the time-history of the horizontal relative displacement between the top of the interior wall and the top of the exterior wall. The maximum horizontal relative displacement was **164.6 mm** at **t = 8.04 s**.

Figure 3.13 shows the time-history of the horizontal relative displacement between the second-floor portion of the interior wall and the second-floor portion of the exterior wall. The maximum horizontal relative displacement was **10.63 mm** at **t = 2.65 s**.

In the two-story RC Upright building, **100 mm × 100 mm L-shaped steel members** are attached to the walls, and the second-floor structure is supported by these steel members. Under the El Centro NS ground motion, the maximum horizontal relative displacement between the second-floor portions of the interior and exterior walls was **10.63 mm**, which is well below **100 mm**. However, since future earthquakes may involve larger amplitudes and longer vibration periods, measures to prevent the collapse or detachment of the second-floor structure should be considered from the standpoint of occupant safety. For example, the second-floor framing could be connected to the L-shaped steel members using chains or other devices with sufficient strength.

Figure 3.14 shows the deformed shape at the instant when the maximum horizontal relative displacement occurred between the exterior wall top and the slab (**12.2 mm at t = 2.93 s**). As shown in the figure, the exterior and interior walls deform **in phase**.

Figure 3.15 shows the deformed shape at the instant when the maximum horizontal relative displacement occurred between the interior wall top and the slab (**26.4 mm at t = 5.02 s**). As shown in the figure, the two interior walls deform **in phase** at **t = 5.02 s**, whereas the exterior and interior walls deform **out of phase**. This behavior results in a large horizontal relative displacement between the tops of the interior and exterior walls.

Furthermore, when the **El Centro NS ground motion** (peak acceleration: **341.7 gal**, peak velocity: **33.5 kine**, peak acceleration response spectrum: approximately **7,000 gal**) was applied to the two-story RC Upright building, the maximum horizontal relative displacement between the exterior wall top and the slab was only about **26 mm**. The analysis indicated that cracking damage in the walls would be extremely minor, and that the reinforcing bars would remain below their yield strength, maintaining structural integrity.

One reason why the building remains structurally sound under the El Centro NS earthquake excitation is its exceptionally high structural capacity. In a conventional low-rise reinforced concrete building, columns are typically about **450 mm × 450 mm** in cross-section. In contrast, the two-story RC Upright building incorporates structural walls whose strength is considered equivalent to approximately **70 reinforced concrete columns measuring 300 mm × 300 mm**. In addition, components other than the main structural system are constructed of lightweight timber, resulting in a relatively small structural load. These factors are considered to contribute significantly to the building's excellent seismic performance.

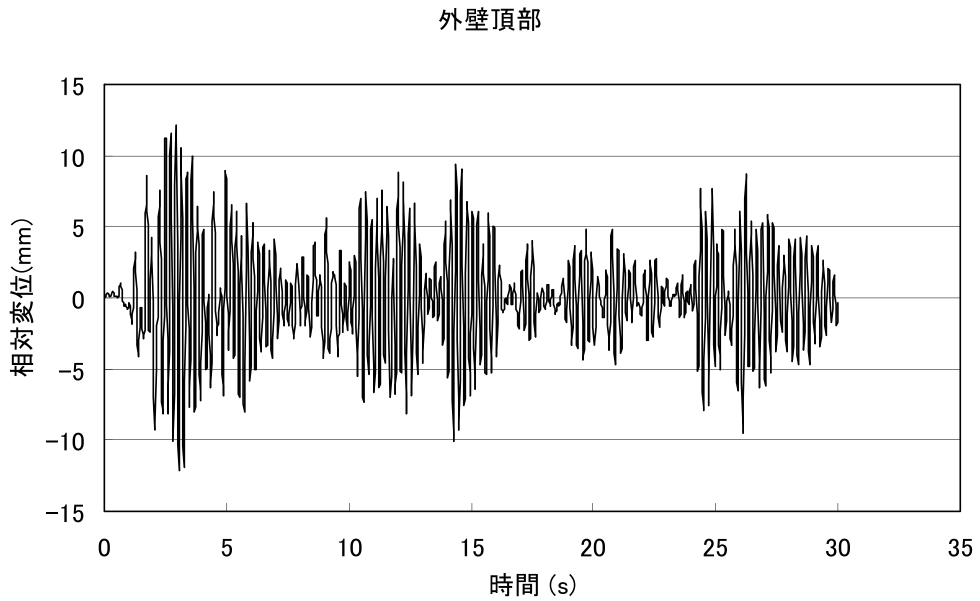


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Figure 3.8 Time History of Horizontal Relative Displacement Between the Top of the Exterior Wall and the Slab (Maximum Relative Displacement = 12.2 mm at t = 2.93 s)

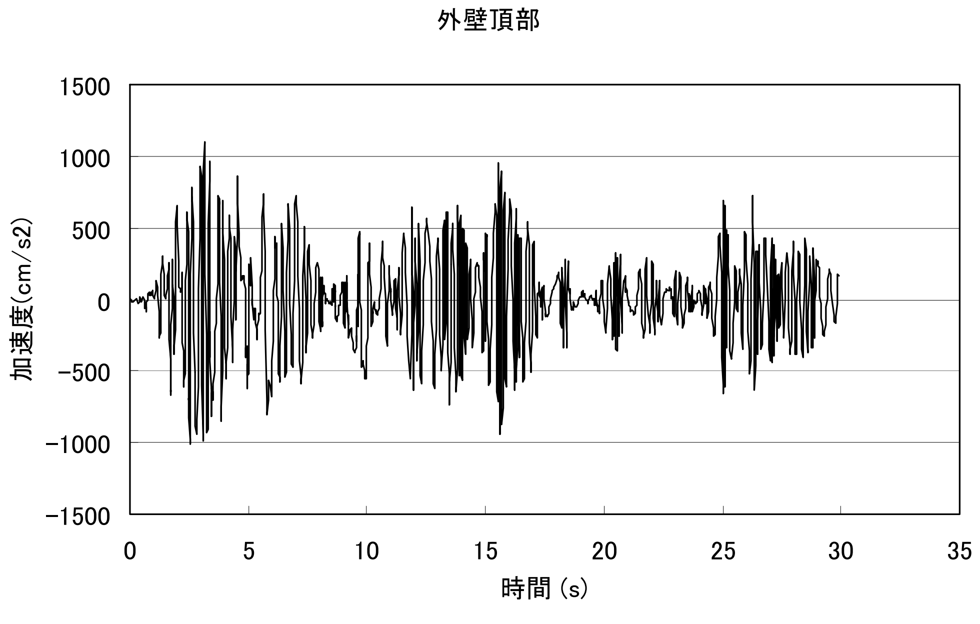

Figure 3.9 Time History of Horizontal Acceleration at the Top of the Exterior Wall (Maximum Acceleration = 1,098 gal at t = 3.14 s)

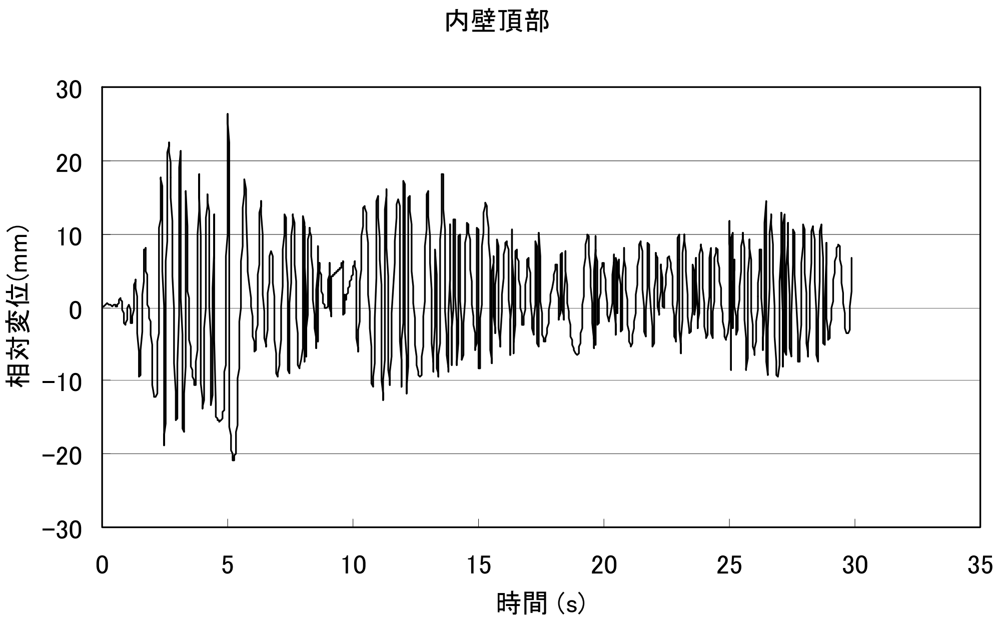

Figure 3.10 Time History of Horizontal Relative Displacement Between the Top of the Interior Wall and the Slab (Maximum Relative Displacement = 26.4 mm at t = 5.02 s)

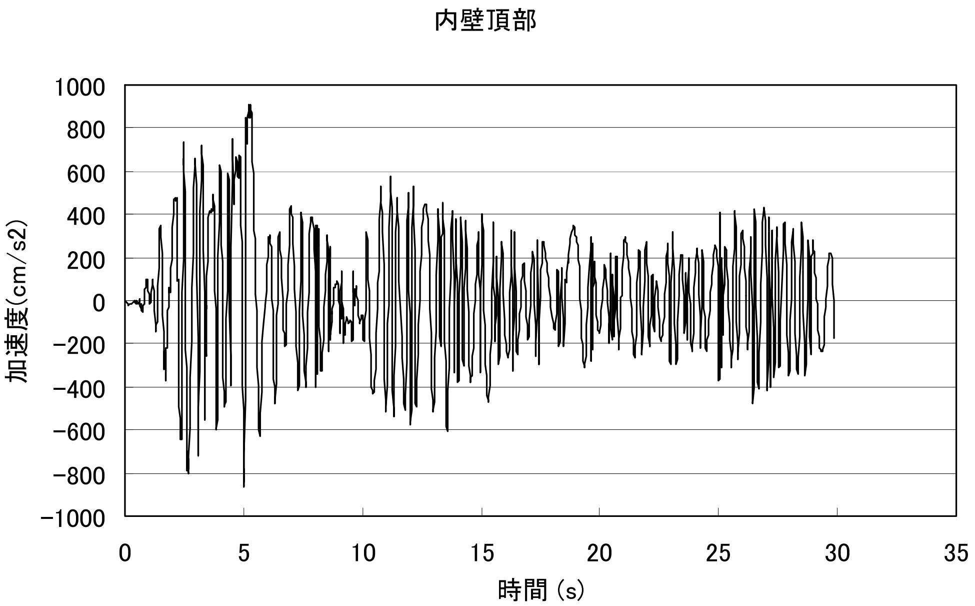

Figure 3.11 Time History of Horizontal Acceleration at the Top of the Interior Wall (Maximum Acceleration = 883 gal at t = 5.31 s)

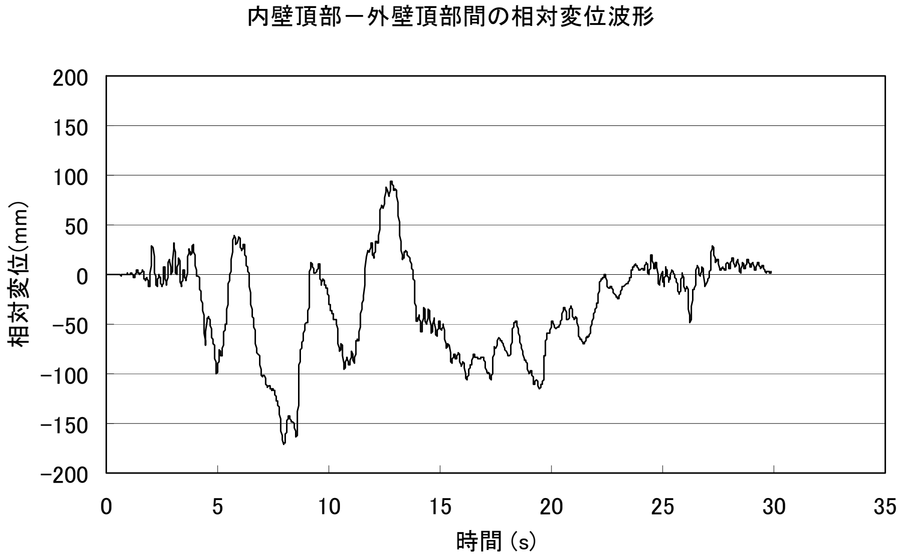

Figure 3.12 Time History of Horizontal Relative Displacement Between the Top of the Interior Wall and the Top of the Exterior Wall (Maximum Relative Displacement = 164.6 mm at t = 8.04 s)

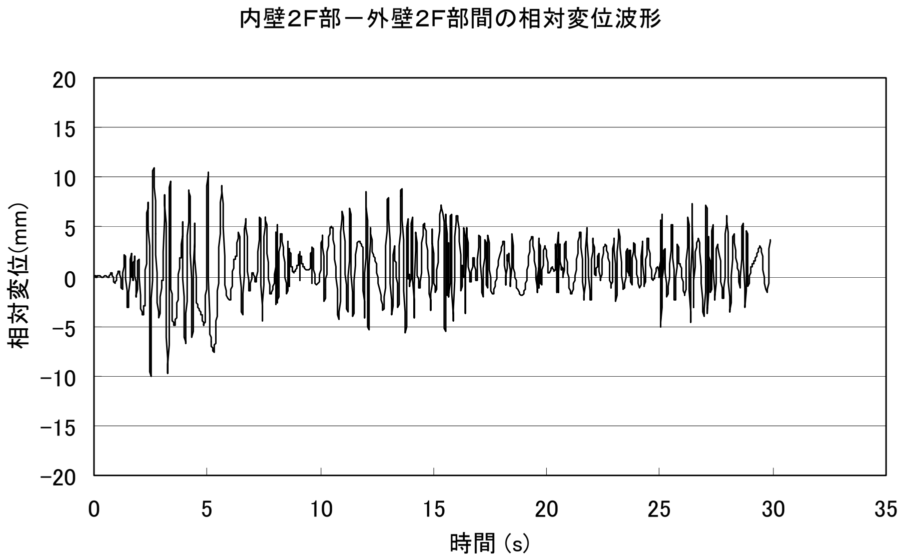

Figure 3.13 Time History of Horizontal Relative Displacement Between the Second-Floor Portions of the Interior Wall and the Exterior Wall (Maximum Relative Displacement = 10.63 mm at t = 2.65 s)

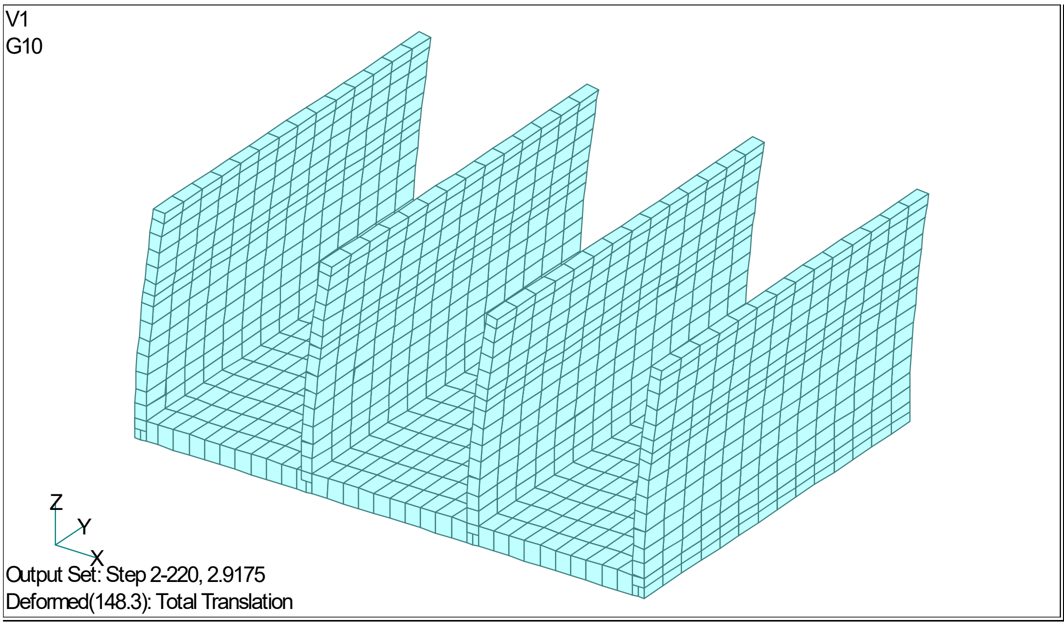

Figure 3.14 Deformed Shape at the Time of Maximum Horizontal Relative Displacement Between the Top of the Exterior Wall and the Slab (Maximum Relative Displacement = 12.2 mm at t = 2.93 s)

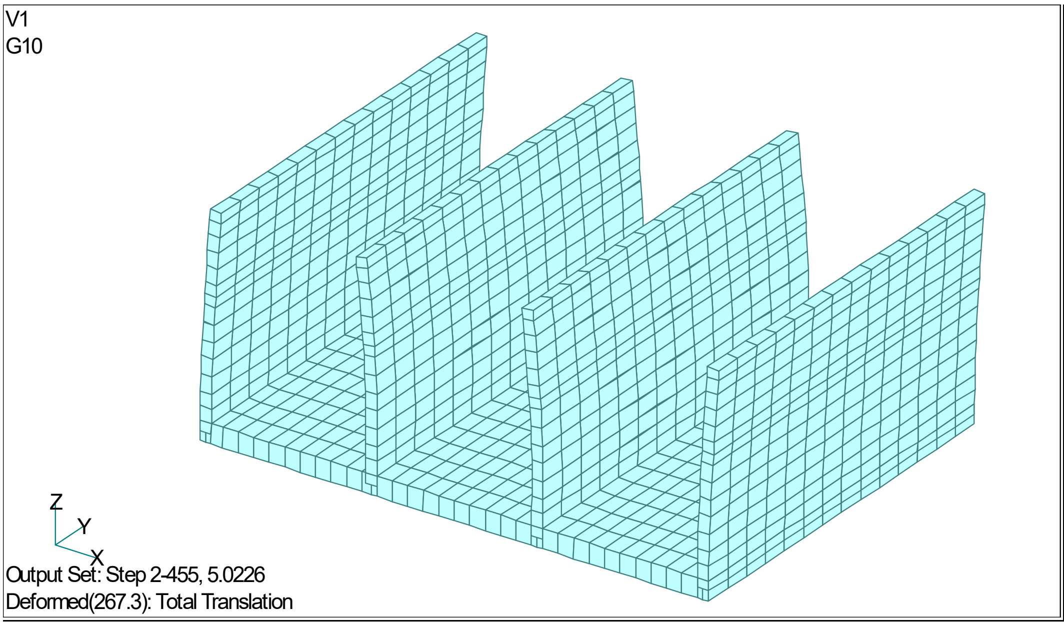

Figure 3.15 Deformed Shape at the Time of Maximum Horizontal Relative Displacement Between the Top of the Interior Wall and the Slab (Maximum Relative Displacement = 26.4 mm at t = 5.02 s)

4. Conclusions

In this study, the two-story building constructed using the **RC Upright Construction Method** was modeled using a **three-dimensional thick-shell finite element model incorporating reinforcing bars**, and a **nonlinear dynamic earthquake response analysis** was performed to evaluate its structural integrity under seismic loading.

The principal findings obtained from this analysis, together with topics requiring further investigation, are summarized below.

(1)

According to the eigenvalue analysis results of the two-story building constructed using the **RC Upright Construction Method**, the **first natural frequency** of the building was found to be **f_1 = 3.13 Hz (T_1 = 0.32 s)**.

In addition, a separate field measurement survey was conducted in which a vibration exciter was installed on the second floor of the building to measure its natural frequency. The measured **first natural frequency** was **f_1 = 6.2 Hz (T_1 = 0.16 s)**, indicating that the building possesses a relatively stiff structural system.

The first natural frequency obtained from the present eigenvalue analysis, **f_1 = 3.13 Hz (T_1 = 0.32 s)**, is lower than the measured value. This discrepancy is considered to be due to the assumptions used in the analytical model. In the present analysis, the stiffness contributions of the roof, wooden walls, and wooden floors were neglected. As a result, the overall stiffness of the analytical model was reduced, leading to a lower predicted first natural frequency compared with the measured value of the actual building.

(2)

When the **El Centro NS ground motion** (peak acceleration: **341.7 gal**, peak velocity: **33.5 kine**, peak acceleration response spectrum: approximately **7,000 gal**) was applied to the two-story building constructed using the **RC Upright Construction Method**, the analysis indicated that the **maximum horizontal relative displacement between the top of the exterior wall and the slab was approximately 13 mm**, which is relatively small.

The results further showed that any cracking damage in the walls would be extremely minor, and that the reinforcing bars would remain below their yield strength, indicating that the structural system would remain sound and undamaged.

One reason why the building maintains its structural integrity under the El Centro NS earthquake excitation is its exceptionally high structural capacity. In a conventional low-rise reinforced concrete building, columns are typically about **450 mm × 450 mm** in cross section. In contrast, the structural wall system of the two-story RC Upright building provides a level of strength equivalent to approximately **seventy reinforced concrete columns measuring 300 mm × 300 mm**. Furthermore, components other than the primary structural system are constructed of lightweight timber, resulting in relatively low gravitational loads acting on the structure. These factors are considered to contribute significantly to the building's excellent seismic performance and structural safety.

(3)

As a subject for future study, the weight reduction of buildings constructed using the **RC Upright Construction Method** should be investigated. One possible approach is to increase the wall thickness at the lower portion of the walls while introducing a tapered wall section, thereby reducing the wall thickness toward the upper portion of the structure. Such a configuration may improve structural efficiency while reducing overall weight.

It is also important to perform **large-deformation elasto-plastic analyses** of RC Upright buildings under progressively increasing horizontal seismic accelerations in order to determine their **ultimate acceleration capacity** and seismic performance limits.

Based on the results of such analyses, it will be important to conduct additional **nonlinear dynamic earthquake response analyses** using stronger ground motions and earthquake records with different frequency characteristics to further evaluate the structural integrity and seismic resilience of RC Upright buildings under a wider range of seismic conditions.

From a construction and safety perspective, measures to prevent the falling or detachment of second-floor structural components during earthquakes are also important. Appropriate connection details and restraint systems should be incorporated to enhance occupant safety during severe seismic events.

Furthermore, the installation of **base-isolation** and **seismic energy dissipation (vibration control) systems** that are effective against the large-scale earthquakes anticipated in the future should be considered. The development of base-isolation and vibration-control devices specifically optimized for the **RC Upright Construction Method** is also regarded as an important area for future research and development.

Appendix 1 Additional Masses Assigned to Walls and Slabs

Source:
Reference (18): Structural Design Calculation Report for the (Tentative) Takashima No. 3 New Construction Project (708), October 2004.

Appendix 2 Representative Recorded Earthquake Motions (Acceleration Data)

Source:
Reference (11): Building Center of Japan (BCJ), "Representative Recorded Earthquake Motions (Acceleration Data)."

Appendix 3 Design Response Spectra for Horizontal Seismic Motions

Source:
Reference (10): Building Research Institute, Ministry of Construction, and Building Center of Japan, *Technical Guidelines (Draft) for the Generation of Design Input Ground Motions – Commentary Edition*, March 1992.

In the design velocity response spectra (Level 1 and Level 2) shown on the following page, a velocity of **100 kine** corresponds approximately to an acceleration response spectrum value of **1,000 gal**. The thick solid line in the figures represents the velocity response spectrum of the design artificial ground motion **B(T)**.

The following pages present the acceleration, velocity, and displacement time histories of the design artificial ground motions **B(T) (Level 1 and Level 2)**. The maximum accelerations of the artificial ground motions B(T) are **207.3 gal (Level 1)** and **355.7 gal (Level 2)**. However, the corresponding design velocity response spectra have maximum values of **50 kine (Level 1, equivalent to approximately 500 gal)** and **100 kine (Level 2, equivalent to approximately 1,000 gal)**, respectively.

The design velocity response spectra (Level 1 and Level 2) also include the velocity response spectra of the **El Centro NS ground motion**, whose amplitudes have been adjusted to **20 kine (Level 1)** and **50 kine (Level 2)**, respectively. These spectra are indicated by dotted lines and yellow markers in the figures.

The El Centro NS ground motion used in the present earthquake response analysis (**341.7 gal, 33.5 kine**) is considered to correspond to an acceleration response spectrum with a peak value of approximately **700 gal**.

For reference, the acceleration response spectra (**h = 5%**) and velocity response spectra (**h = 5%**) of the **1995 Hyogo-ken Nanbu (Great Hanshin) Earthquake** and the **2004 Niigata Chuetsu Earthquake** are also presented, based on data published on the website of the **Disaster Prevention Research Institute, Kyoto University (Prof. Iwata)**.

As shown in the figures, the **Niigata Chuetsu Earthquake** exhibited an extremely large seismic intensity, with a maximum acceleration response spectrum of approximately **6,000 gal (h = 5%)** and a maximum velocity response spectrum of approximately **600 kine (h = 5%)**, which are significantly greater than those observed during the **Hyogo-ken Nanbu Earthquake**.

Appendix 4 ABAQUS Concrete Model

Source:

Reference (19): ABAQUS Ver. 6.5 User's Manual, ABAQUS Theory Manual.

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